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Series Bjournal homepage: www.elsevier.com/locate/jctbClustered coloring of graphs excluding a subgraph
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ABSTRACT

A graph coloring has bounded clustering if each monochromatic component has bounded size. Equivalently, it is a partition of the vertices into induced subgraphs with bounded size components. This paper studies clustered colorings of graphs, where the number of colors depends on an excluded minor and/or an excluded subgraph. We prove the following results (for fixed integers s, t and a fixed graph H). First we show that graphs with no $K_{s,t}$ subgraph and with no H -minor are $(s+2)$ -colorable with bounded clustering. The number of colors here is best possible. This result implies that graphs with no K_{s+1} -minor are $(s+2)$ -colorable with bounded clustering, which is within two colors of the clustered coloring version of Hadwiger's conjecture. For graphs of bounded treewidth (or equivalently, excluding a planar minor) and with no $K_{s,t}$ subgraph, we prove $(s+1)$ -choosability with bounded clustering, which is best possible. We then consider excluding an odd minor. We prove that graphs with no $K_{s,t}$ subgraph and with no odd H -minor are $(2s+1)$ -colorable with bounded clustering, generalizing a result of the first author and Oum who proved the case $s = 1$. Moreover, at least $s-1$ color classes are stable sets. Finally, we consider the clustered coloring version of a conjecture of Gerards and Seymour and prove

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that graphs with no odd K_{s+1} -minor are $(8s - 4)$ -colorable with bounded clustering, which improves on previous such bounds.

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1. Introduction

Hadwiger's conjecture [19] asserts that every graph with no K_{s+1} -minor has a proper s -coloring. For $s \leq 2$ the conjecture is easy. Hadwiger [19] and Dirac [8] independently proved the $s = 3$ case. Wagner [57] proved that Hadwiger's conjecture with $s = 4$ is equivalent to the Four Color Theorem [45]. And Robertson, Seymour, and Thomas [50] proved Hadwiger's conjecture for $s = 5$. The conjecture remains open for $s \geq 6$. Hadwiger's conjecture is widely considered to be one of the most important open problems in graph theory. The best known upper bound on the chromatic number of K_{s+1} -minor-free graphs is $O(s \log \log s)$ due to Delcourt and Postle [7], improving a recent breakthrough of Norin et al. [38] who improved a long-standing bound independently due to Kostochka [29,30] and Thomason [54,55]. Indeed, it is open whether every graph with no K_{s+1} -minor is $O(s)$ -colorable. See the recent survey by Seymour [52] for more on Hadwiger's conjecture.

One way to approach Hadwiger's conjecture is to allow improper colorings. Say that a *coloring* of a graph G is simply a function that assigns one color to each vertex of G . A *monochromatic component* with respect to a coloring of G is a connected component of the subgraph of G induced by all the vertices assigned a single color. A coloring has *clustering* η if every monochromatic component has at most η vertices. The *clustered chromatic number* of a graph class $\mathcal{G}$ is the minimum integer k for which there exists an integer η such that every graph in $\mathcal{G}$ is k -colorable with clustering η . There have been several recent papers on this topic [1,4,12,15,16,21–23,25,27,28,31,32,35,39]; see [59] for a survey.

Kawarabayashi and Mohar [28] first proved a $O(s)$ upper bound on the clustered chromatic number of K_{s+1} -minor-free graphs. The number of colors has since been steadily improved, as shown in Table 1, where $\eta(s)$ is some large unspecified function.

It remains open whether graphs with no K_{s+1} minor are s -colorable with bounded clustering.³ Note that s colors would be best possible for any fixed clustering value. That is, for all $s \geq 2$ and η there is a graph G with no K_{s+1} minor such that every

³ Dvořák and Norin [12] have announced that a forthcoming paper proves that graphs with no K_{s+1} minor are s -colorable, in fact s -choosable, with bounded clustering.

Table 1
Clustered coloring of K_{s+1} -minor-free graphs.

	number of colors	clustering	choosability
Kawarabayashi and Mohar [28]	$\lceil \frac{31}{2}(s+1) \rceil$	$\eta(s)$	yes
Wood [58] ¹	$\lceil \frac{7s+4}{2} \rceil$	$\eta(s)$	yes
Edwards et al. [13]	$4s$	$\eta(s)$	
Liu and Oum [32]	$3s$	$\eta(s)$	
Norin [37] ²	$2s$	$\eta(s)$	
Van den Heuvel and Wood [23]	$2s$	$\lceil \frac{s-2}{2} \rceil$	
Dvořák and Norin [12]	$2s$	$\eta(s)$	

¹ This result depended on a result announced by Norine and Thomas [41,53] which has not yet been written.

² See [52] for some of the details.

$(s-1)$ -coloring of G has a monochromatic component with more than η vertices.⁴ In the following discussion we postpone giving standard definitions until Section 1.8.

1.1. Main results

The current best known bound on the clustered chromatic number of K_{s+1} -minor-free graphs is $2s$ [12,23,37]. We prove the following bound, which is within two colors of best possible.

Theorem 1. *For every $s \in \mathbb{N}$, there exists $\eta \in \mathbb{N}$ such that every graph with no K_{s+1} -minor is $(s+2)$ -colorable with clustering η .*

We in fact prove the following stronger result where the number of colors only depends on an excluded $K_{s,t}$ subgraph. Indeed, the number of colors only depends on s . The dependence on t and the excluded minor is hidden in the clustering function.

Theorem 2. *For all $s, t \in \mathbb{N}$ and for every graph H , there exists $\eta \in \mathbb{N}$ such that every graph with no H minor and with no $K_{s,t}$ subgraph is $(s+2)$ -colorable with clustering η .*

Theorem 2 will be proved in Section 5.

Since every graph with no K_{s+1} minor has no $K_{s,s}$ subgraph, Theorem 1 is an immediate corollary of Theorem 2. This theorem and those in our companion papers [33,34] are the first known results for clustered coloring where the number of colors depends on an excluded subgraph. While Theorem 1 is of substantial interest, we emphasise that our main results are for graph classes excluding a $K_{s,t}$ subgraph. One motivation for this line of research is that a graph contains no $K_{1,t}$ subgraph if and only if it has maximum degree less than t . So Theorem 2 generalizes a result by the first author and Oum [32]

⁴ Edwards et al. [13] proved the following stronger lower bound: for all $s \geq 2$ and c there is a graph G with no K_{s+1} minor such that every $(s-1)$ -coloring of G has a monochromatic component with maximum degree greater than c . Conversely, Edwards et al. [13] proved that every graph with no K_{s+1} minor is s -colorable such that each monochromatic component has maximum degree $O(s^2 \log s)$. This degree bound was improved to $O(s)$ by van den Heuvel and the second author [23].

who proved the $s = 1$ case which was originally conjectured by Esperet and Joret [15]. Also note that excluding a non-forest subgraph alone is not enough to guarantee bounded clustered chromatic number. In particular, for every graph H that contains a cycle, and for all $k, \eta \in \mathbb{N}$, if G is a graph with chromatic number greater than $k\eta$ and girth greater than $|V(H)|$ (which exists [14]), then G contains no H subgraph and G is not k -colorable with clustering η , for otherwise G would be $k\eta$ -colorable.

While Theorem 1 is within two colors of the conjectured answer, we now show that the number of colors in Theorem 2 is best possible. The proof is a variation on the well known “standard” example; see [59]. We claim that for all $s, \eta \in \mathbb{N}$ there is a graph G_s with no K_{s+4} minor and with no $K_{s,s+6}$ subgraph, such that every $(s+1)$ -coloring of G_s has a monochromatic component on at least η vertices. We proceed by induction on s . In the base case, $s = 1$, let G_1 be the $\eta \times \eta$ triangular grid graph. Then G_1 has no K_5 minor since it is planar, and G_1 has no $K_{1,7}$ subgraph since it has maximum degree 6. By the Hex Lemma [17], every 2-coloring of G_1 has a monochromatic path on η vertices, as claimed. Now assume the claim for G_{s-1} . Let G_s be obtained from η disjoint copies of G_{s-1} by adding a new vertex v adjacent to all other vertices. Each component of $G_s - v$ is a copy of G_{s-1} . If G_s contains a K_{s+4} minor, then some component of $G_s - v$ contains a K_{s+3} minor, which is a contradiction. Thus G_s contains no K_{s+4} minor. Similarly, if G_s contains a $K_{s,s+6}$ subgraph, then $G_s - v$ contains a $K_{s-1,s+6}$ or $K_{s,s+5}$ subgraph, both of which contain $K_{s-1,(s-1)+6}$, which is a contradiction. Thus G_s contains no $K_{s,s+6}$ subgraph. Now consider an $(s+1)$ -coloring of G_s . Say v is blue. If every component of $G_s - v$ has a blue vertex, then the blue subgraph contains a star on $\eta + 1$ vertices, and we are done. Otherwise, some component X of $G_s - v$ has no blue vertex, and thus has only s colors. By induction, X and hence G_s contains a monochromatic component with at least η vertices, as desired.

1.2. Colin de Verdière parameter

The Colin de Verdière parameter $\mu(G)$ is an important minor-closed graph invariant introduced by Colin de Verdière [5,6]; see [51,56] for surveys. It is known that $\mu(G) \leq 1$ if and only if G is a union of disjoint paths, $\mu(G) \leq 2$ if and only if G is outerplanar, $\mu(G) \leq 3$ if and only if G is planar, and $\mu(G) \leq 4$ if and only if G is linklessly embeddable. A famous conjecture of Colin de Verdière [5] asserts that every graph G with $\mu(G) \leq s$ is properly $(s+1)$ -colorable. This implies the Four Color Theorem, and is implied by Hadwiger’s conjecture. It is open whether every graph G with $\mu(G) \leq s$ is $(s+1)$ -colorable with bounded clustering. Every graph G with $\mu(G) \leq s$ contains no K_{s+2} minor. So Theorem 1 implies that such graphs are $(s+3)$ -colorable with bounded clustering, but a better bound can be obtained by Theorem 2. Van der Holst, Lovász and Schrijver [56] proved that $\mu(K_{s,t}) = s+1$ for $t \geq \max\{s, 3\}$. Thus if $\mu(G) \leq s$ then G contains no $K_{s,t}$ subgraph (since μ is monotone under taking subgraphs). Theorem 2 then implies:

Corollary 3. *For every $s \in \mathbb{N}$ there exists $\eta \in \mathbb{N}$, such that every graph G with $\mu(G) \leq s$ is $(s + 2)$ -colorable with clustering η .*

This example highlights the utility of excluding a subgraph within a minor-closed class.

1.3. Bounded treewidth

When the excluded minor H is planar (or equivalently, when the graph has bounded treewidth), Theorem 2 is improved as follows.

Theorem 4. *For all $s, t, w \in \mathbb{N}$ there exists $\eta \in \mathbb{N}$ such that every graph with treewidth at most w and with no $K_{s,t}$ subgraph is $(s + 1)$ -colorable with clustering η .*

The number of colors in Theorem 4 is best possible: for all $s, c \in \mathbb{N}$ there is a graph G with treewidth s , with no $K_{s,s+2}$ subgraph, and such that every s -coloring of G has a monochromatic component on at least c vertices. The construction is analogous to the construction above except that in the base case ($s = 1$) we use a long path instead of the triangular grid. This is called a “standard” example in [59].

We actually prove the following list-coloring result, which immediately implies Theorem 4.

Theorem 5. *For all $s, t, w \in \mathbb{N}$, there exists $\eta \in \mathbb{N}$ such that every graph with treewidth at most w and with no $K_{s,t}$ subgraph is $(s + 1)$ -choosable with clustering η .*

Theorem 5 will be proved in Section 3.

The case $s = 1$ of Theorem 5 is an unpublished result of the first author (see [52, Theorem 6.4]), which generalizes a result of Alon, Ding, Oporowski, and Vertigan [1] who proved Theorem 4 in the case $s = 1$ (with much better bounds on η).

Theorem 5 immediately implies results for graphs with bounded treewidth and with no $K_{s,t}$ -minor, although we emphasise that Theorem 5 holds in the stronger setting of an excluded $K_{s,t}$ subgraph. In particular, Theorem 5 implies that graphs with bounded treewidth and with no $K_{s,t}$ minor are $(s + 1)$ -choosable with bounded clustering. Since K_{s+1} is a minor of $K_{s,s}$, this in turn implies that graphs with bounded treewidth and with no K_{s+1} -minor are $(s + 1)$ -choosable with bounded clustering. Dvořák and Norin [12] proved this result with one fewer color. That is, graphs with bounded treewidth and with no K_{s+1} -minor are s -choosable with bounded clustering. In our companion paper [34], we strengthen this result by showing that graphs with bounded treewidth and with no K_{s+1} -topological-minor are s -choosable with bounded clustering. This says that a clustered version of Hajós’ conjecture holds for bounded treewidth graphs, and even holds for choosability. Results in this paper are critical components for [34].

1.4. Excluded odd minors

Gerards and Seymour (see [24, §6.5]) conjectured that every graph with no odd K_{s+1} minor is properly s -colorable, which implies Hadwiger's conjecture. The best known upper bound on the chromatic number of graphs with no odd K_{s+1} minor is $O(s(\log \log s)^6)$, due to Postle [43], improving earlier results of Geelen, Gerards, Reed, Seymour, and Vetta [18] and Norin and Song [40]. It is open whether such graphs are properly $O(s)$ colorable. The first $O(s)$ bound on the clustered chromatic number was established by Kawarabayashi [27], who proved that every graph with no odd K_s minor is $496s$ -colorable with bounded clustering. The number of colors was improved to $10s - 13$ by Kang and Oum [25]. We make the following modest improvement.

Theorem 6. *For all $s \in \mathbb{N}$ there exists $\eta \in \mathbb{N}$ such that every graph with no odd K_{s+1} minor is $(8s - 4)$ -colorable with clustering η .*

Theorem 6 will be proved in Section 6.

More interestingly, we prove the following analogue of Theorem 2 for excluded odd minors and excluded subgraphs.

Theorem 7. *For all $s, t \in \mathbb{N}$ and for every graph H there exists $\eta \in \mathbb{N}$ such that every graph with no odd H -minor and with no $K_{s,t}$ subgraph is $(2s+1)$ -colorable with clustering η . Moreover, at least $s - 1$ color classes are stable sets.*

Theorem 7 will be proved in Section 5.

The case of $s = 1$ in Theorem 7 was proved by the first author and Oum [32]. Here, three colors is best possible.

Note that no clustered choosability result is possible for graphs excluding an odd minor, since the complete bipartite graph $K_{n,n}$ contains no odd K_3 minor, but it follows from the work of Kang [26] that for all $k, \eta \in \mathbb{N}$ there exists $n \in \mathbb{N}$ such that $K_{n,n}$ is not k -choosable with clustering η .

1.5. $K_{s,t}$ -minor-free graphs

Consider graphs with no $K_{s,t}$ minor for $s \leq t$. Van den Heuvel and the second author [23] observed that results of Edwards et al. [13] and Ossona de Mendez, Oum and the second author [42] imply that such graphs are $3s$ -colorable with bounded clustering, which was improved to $2s + 2$ by Dvořák and Norin [12]. Theorem 2 immediately implies the following further improvement:

Corollary 8. *For all $s, t \in \mathbb{N}$, there exists $\eta \in \mathbb{N}$ such that every graph with no $K_{s,t}$ -minor is $(s + 2)$ -colorable with clustering η .*

The best known lower bound on the clustered chromatic number of $K_{s,t}$ -minor-free graphs is $s + 1$, due to van den Heuvel and the second author [23]. It is open whether every $K_{s,t}$ -minor-free graph is $(s + 1)$ -colorable with bounded clustering. Van den Heuvel and the second author [23] proved this in the $s = 2$ case.

1.6. H -minor-free graphs

Now consider the clustered chromatic number of H -minor-free graphs, for an arbitrary graph H . A *vertex cover* of a graph H is a set $S \subseteq V(H)$ such that $H - S$ has no edges. Suppose that H has a vertex cover of size s . Then H is a minor of $K_{s,|V(H)|-1}$ (obtained by contracting a matching of size $s - 1$ in $K_{s,|V(H)|-1}$). So every graph containing no H -minor contains no $K_{s,|V(H)|-1}$ -minor. Corollary 8 thus implies:

Corollary 9. *For all $s \in \mathbb{N}$ and for every graph H that has a vertex-cover of size at most s , there exists $\eta \in \mathbb{N}$ such that every graph with no H -minor is $(s + 2)$ -colorable with clustering η .*

We now relate this result to a conjecture of Norin, Scott, Seymour and the second author [39] about the clustered chromatic number of H -minor-free graphs. Let T be a rooted tree. The *depth* of T is the maximum number of vertices on a root-to-leaf path in T . The *closure* of T is obtained from T by adding an edge between every ancestor and descendent in T . The *connected tree-depth* of a graph H , denoted by $\overline{\text{td}}(H)$, is the minimum depth of a rooted tree T such that H is a subgraph of the closure of T . Norin et al. [39] observed that for every graph H and $\eta \in \mathbb{N}$ there is an H -minor-free graph that is not $(\overline{\text{td}}(H) - 2)$ -colorable with clustering η ; thus the clustered chromatic number of H -minor-free graphs is at least $\overline{\text{td}}(H) - 1$. On the other hand, Norin et al. [39] conjectured that the class of H -minor-free graphs has clustered chromatic number at most $2\overline{\text{td}}(H) - 2$, which would be tight for certain graphs H . As evidence for this conjecture, Norin et al. [39] proved that the clustered chromatic number of H -minor-free graphs is at most $2^{\overline{\text{td}}(H)+1} - 4$.

For $h \in \mathbb{N}$ with $h \geq 3$, a *broom* of height h is a rooted tree that can be obtained from a star rooted at a leaf by subdividing the edge incident with the root $h - 3$ times. The closure of a broom of height h has a vertex-cover of size at most $h - 1$. Thus Corollary 9 can be restated as follows:

Corollary 10. *For every integer $h \geq 3$, if H is a subgraph of the closure of the broom of height h , then there exists $\eta \in \mathbb{N}$ such that every graph with no H -minor is $(h + 1)$ -colorable with clustering η .*

Note that the depth of the broom of height h is h . So Corollary 10 answers the aforementioned conjecture of Norin et al. [39] in a stronger sense (since $h + 1 \leq 2h - 2$) when the underlying tree T is a broom.

1.7. Excluded subdivisions

Our companion paper [34] studies clustered colorings of graphs excluding various graph subdivisions. The methods build heavily on those introduced in this paper and reference [34] uses some results in this paper as a black box. For example, we prove in [34] that graphs with bounded treewidth and with no almost (≤ 1) -subdivision of K_{s+1} are s -choosable with bounded clustering. Here a graph is an *almost (≤ 1) -subdivision* of a graph H if it can be obtained from H by subdividing edges, where at most one edge is subdivided more than once. This result is a clustered choosability version of Hajós conjecture for graphs of bounded treewidth. Allowing one more color, we prove an analogous result for graphs excluding a fixed minor; that is, we prove that graphs excluding a fixed graph as a minor and with no almost (≤ 1) -subdivision of K_{s+1} are $(s+1)$ -colorable with bounded clustering.

1.8. Standard definitions

Let $\mathbb{N} := \{1, 2, \dots\}$ and $\mathbb{N}_0 := \{0, 1, 2, \dots\}$. For $m, n \in \mathbb{N}$, let $[m, n] := \{m, m+1, \dots, n\}$ and $[n] := [1, n]$.

Let G be a graph (allowing loops and parallel edges). For $v \in V(G)$, let $N_G(v) := \{w \in V(G) : vw \in E(G)\}$ be the neighborhood of v , and let $N_G[v] := N_G(v) \cup \{v\}$. For $X \subseteq V(G)$, let $N_G(X) := \bigcup_{v \in X} (N_G(v) - X)$ and $N_G[X] := N_G(X) \cup X$. Denote the subgraph of G induced by X by $G[X]$.

For our purposes, a *color* is an element of $\mathbb{Z}$. A *list-assignment* of a graph G is a function L with domain containing $V(G)$, such that $L(v)$ is a non-empty set of colors for each vertex $v \in V(G)$. For a list-assignment L of $V(G)$, an L -*coloring* of G is a function c with domain $V(G)$ such that $c(v) \in L(v)$ for every $v \in V(G)$. An L -coloring has *clustering* η if every monochromatic component has at most η vertices. A list-assignment L of a graph G is an ℓ -*list-assignment* if $|L(v)| \geq \ell$ for every vertex $v \in V(G)$. A graph is ℓ -*choosable with clustering* η if G is L -colorable with clustering η for every ℓ -list-assignment L of G .

A graph H is a *minor* of a graph G if a graph isomorphic to H can be obtained from a subgraph of G by contracting edges. Here we allow H to have loops and parallel edges. The following is an alternative definition of graph minor. Let H be a graph. An H -*minor* of a graph G is a map α with domain $V(H) \cup E(H)$ such that:

- For every $h \in V(H)$, $\alpha(h)$ is a nonempty connected subgraph of G (called a *branch set*),
- If h_1 and h_2 are different vertices of H , then $\alpha(h_1)$ and $\alpha(h_2)$ are disjoint.
- For each edge $e = h_1h_2$ of H , $\alpha(e)$ is an edge of G with one end in $\alpha(h_1)$ and one end in $\alpha(h_2)$; furthermore, if $h_1 = h_2$, then $\alpha(e) \in E(G) - E(\alpha(h_1))$.
- If e_1, e_2 are distinct edges of H , then $\alpha(e_1) \neq \alpha(e_2)$.

Then α is an *odd H -minor* if there exists a 2-coloring c of $\bigcup_{h \in V(H)} \alpha(h)$ such that $c|_{\alpha(h)}$ is a proper 2-coloring of $\alpha(h)$, and for every edge e of H , the ends of $\alpha(e)$ receive the same color in c . See [18,25,27,40,43] for work on odd minors.

A *tree decomposition* of a graph G is a pair $(T, \mathcal{X} = (X_x : x \in V(T)))$, where T is a tree and for each node $x \in V(T)$, X_x is a subset of $V(G)$ called a *bag*, such that for each vertex $v \in V(G)$, the set $\{x \in V(T) : v \in X_x\}$ induces a non-empty (connected) subtree of T , and for each edge $vw \in E(G)$ there is a node $x \in V(T)$ such that $v, w \in X_x$. A *path decomposition* is a tree decomposition whose underlying tree is a path. The *width* of a tree decomposition $(T, \mathcal{X})$ is $\max\{|X_x| - 1 : x \in V(T)\}$. The *treewidth* of a graph G is the minimum width of a tree decomposition of G . For each integer k , the graphs with treewidth at most k form a minor-closed class. Robertson and Seymour [46] proved that a minor-closed class of graphs has bounded treewidth if and only if some planar graph is not in the class. treewidth is a key parameter in algorithmic and structural graph theory; see [3,20,44] for surveys.

A *separation* of a graph G is an ordered pair (A, B) of edge-disjoint subgraphs of G with $A \cup B = G$. The *order* of (A, B) is $|V(A \cap B)|$.

A *tangle* $\mathcal{T}$ in a graph G of order $\theta \in \mathbb{Z}$ is a set of separations of G of order less than θ such that the following hold:

- (T1) For every separation (A, B) of G of order less than θ , either $(A, B) \in \mathcal{T}$ or $(B, A) \in \mathcal{T}$.
- (T2) If $(A_i, B_i) \in \mathcal{T}$ for $i \in [3]$, then $A_1 \cup A_2 \cup A_3 \neq G$.
- (T3) If $(A, B) \in \mathcal{T}$, then $V(A) \neq V(G)$.

A *surface* is a nonnull compact connected 2-manifold without boundary. Every surface is homeomorphic to the sphere with k handles (which has Euler genus $2k$) or the sphere with k cross-caps (which has Euler genus k). The *Euler genus* of a graph G is the minimum Euler genus of a surface in which G embeds; see [36] for more on graph embeddings.

2. List coloring setup

We prove Theorems 2, 5 and 7 using the same technique. A key is to actually prove stronger results that allow for a bounded-size set Y of precolored vertices. We then require that not only every monochromatic component has bounded size, but also that the union of all the monochromatic components intersecting Y has size at most $g(|Y|)$, for some function g . Assume that G is a minimum counterexample, and subject to this, the size of Y is as large as possible. We distinguish two cases depending on the size of Y .

First consider the case that Y is large. Let θ be a large number. If there exists a separation (A, B) of G of order less than θ such that both $V(A) \cap Y$ and $V(B) \cap Y$ contains at least 3θ vertices, then we can precolor the vertices in $V(A \cap B)$ so that

the number of precolored vertices in A (and B , respectively) is smaller than $|Y|$, apply induction to each of A and B with the new precolored set to obtain a coloring of A and a coloring of B , and then combine the colorings to obtain a coloring of G . So we may assume that such a separation does not exist. This defines a tangle of order θ . But such a tangle does not exist when the graph has bounded treewidth, which finishes this case for graphs of bounded treewidth (Theorem 5).

For graphs excluding a minor (which might have unbounded treewidth), we apply Robertson and Seymour's Graph Minor Structure Theorem [49], which describes the structure of graphs excluding a minor relative to a tangle of large order. For graphs excluding an odd minor, the extra ingredient is the structure theorem of Geelen, Gerards, Reed, Seymour, and Vetta [18]. We then apply a result of Dujmović, Morin, and the second author [11], from which we (roughly) conclude that ignoring the apex vertices, our graph has bounded layered treewidth (defined below). We then apply a result in our companion paper [33] that shows $(s+2)$ -colorability with bounded clustering for graphs of bounded layered treewidth with no $K_{s,t}$ -subgraph. From this we conclude the result.

It remains to deal with the case that Y is small.

For the time being, assume that $s = 1$; that is, G has bounded maximum degree. Let $c_1, c_2, \dots, c_r$ be the colors appearing in Y , where $r \leq |Y|$. First precolor $N(Y)$ so that no vertex uses c_1 , then precolor $N(N(Y))$ so that no vertex uses c_2 , and so on, until the r -th neighborhood of Y is precolored so that no vertex uses c_r . Let Y' be the set of vertices at distance at most r from Y . Since Y is small and G has bounded maximum degree, $|Y'|$ is bounded. Apply induction to obtain a desired coloring of G with Y' precolored. Note that in this coloring of G , every monochromatic component intersecting Y is contained in Y' so the size is bounded by $g(|Y|)$.

However, this approach for enlarging Y does not work directly when $s \geq 2$, since the precolored set might grow too fast when the maximum degree is unbounded. We employ the following alternative strategy. Instead of precoloring every vertex that is adjacent to the currently precolored set, only precolor those vertices that are adjacent to at least s currently precolored vertices so that they forbid one color in Y , and for each vertex v that is adjacent to at least 1 but at most $s-1$ currently precolored vertices, we ensure (using a list coloring argument) that in the future v is assigned a color that appears on no precolored neighbor of v . This allows us to enlarge Y to obtain a larger precolored set Y' , such that in every coloring, every monochromatic component that intersects Y is contained in Y' , so it has size less than $g(|Y|)$. Lemma 11 below, which is proved in our companion paper [33], ensures that the size of the precolored set does not increase too much, which is then used to ensure that the final precolored set Y' has bounded size.

The following definitions formalise these ideas. For a graph G , a set $X \subseteq V(G)$ and $s \in \mathbb{N}$, define

$$N_G^{\geq s}(X) := \{v \in V(G) - X : |N_G(v) \cap X| \geq s\} \text{ and}$$

$$N_G^{< s}(X) := \{v \in V(G) - X : 1 \leq |N_G(v) \cap X| < s\}.$$

When the graph G is clear from the context we write $N^{\geq s}(X)$ instead of $N_G^{\geq s}(X)$, and similarly for $N^{< s}(X)$.

Lemma 11 ([33]). *For all $s, t \in \mathbb{N}$, there exists a function $f_{s,t} : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ such that for every graph G with no $K_{s,t}$ subgraph, if $X \subseteq V(G)$ then $|N^{\geq s}(X)| \leq f_{s,t}(|X|)$.*

When G excludes a fixed minor, the function $f_{s,t}$ in Lemma 11 can be made linear; see [33]. This improves the clustering function in all our results, but for ease of presentation we choose not to evaluate explicit clustering functions in this paper.

All our results rely on the following list coloring setup. For $s, r \in \mathbb{N}$ and $Y_1 \subseteq V(G)$, a list-assignment L of a graph G is an (s, r, Y_1) -list-assignment if the following hold:

- (L1) $|L(v)| \in [s + r]$ for every $v \in V(G)$.
- (L2) $Y_1 = \{v \in V(G) : |L(v)| = 1\}$.
- (L3) For every $y \in N^{< s}(Y_1)$,

$$|L(y)| = s + r - |N_G(y) \cap Y_1|,$$

and $L(y) \cap L(u) = \emptyset$ for every $u \in N_G(y) \cap Y_1$. (Note that $|L(y)| \geq r + 1$.)

- (L4) For every $v \in V(G) - N_G[Y_1]$, we have $|L(v)| = s + r$.
- (L5) For every $v \in V(G) - Y_1$, we have $|L(v)| \geq r + 1$.

We use $r = 1$ for the bounded treewidth case (Theorem 5), $r = 2$ for excluded minors (Theorem 2), and $r = s + 1$ for excluded odd minors (Theorem 7). Define an (s, Y_1) -list-assignment to be an $(s, 1, Y_1)$ -assignment.

Let G be a graph, let $Y_1 \subseteq V(G)$, let $s, r \in \mathbb{N}$, and let L be an (s, r, Y_1) -list-assignment. For all $W \subseteq V(G)$ and for every set F of colors with $|F| \leq r$ (not necessarily a subset of $\bigcup_{v \in V(G)} L(v)$), a (W, F) -progress of L is a list-assignment L' of G defined as follows:

- Let $Y'_1 := Y_1 \cup W$.
- For every $y \in Y_1$, let $L'(y) := L(y)$.
- For every $y \in Y'_1 - Y_1$, let $L'(y)$ be a 1-element subset of $L(y) - F$ (which exists by (L5)).
- For each $v \in N^{< s}(Y'_1)$, let $L'(v)$ be a subset of

$$L(v) - \bigcup_{w \in N_G(v) \cap (W - Y_1)} L'(w)$$

of size $|L(v)| - |N_G(v) \cap (W - Y_1)|$ such that $|L'(v) \cap F|$ is as large as possible.

- For every $v \in V(G) - (Y'_1 \cup N^{< s}(Y'_1))$, let $L'(v) := L(v)$.

Intuitively speaking, the (W, F) -progress is a list assignment where each uncolored vertex in W is assigned a color in its list but not in F .

Lemma 12. Let G be a graph, $s, r \in \mathbb{N}$, $Y_1 \subseteq V(G)$, and L be an (s, r, Y_1) -list-assignment. If $W \subseteq V(G)$ and F is a set of colors with $|F| \leq r$, then every (W, F) -progress L' of L satisfies the following properties:

1. L' is an $(s, r, Y_1 \cup W)$ -list-assignment of G .
2. $L'(v) \subseteq L(v)$ for every $v \in V(G)$.
3. $\{v \in Y_1 \cup W : L'(v) \cap F \neq \emptyset\} = \{v \in Y_1 : L(v) \cap F \neq \emptyset\}$.
4. If $N^{\geq s}(Y_1) \subseteq W$, then for every $y \in Y_1 \cup W$ and color $x \in F \cap L'(y)$, we have $\{v \in N_G(y) - (Y_1 \cup W) : x \in L'(v)\} = \emptyset$.
5. For every $v \in V(G) - (Y_1 \cup W)$, we have $L'(v) \cap F = L(v) \cap F$.

Proof. Let L' be a (W, F) -progress of L . By construction, Statements 2 and 3 hold. Let $Y'_1 := Y_1 \cup W$.

Now we prove Statement 4. Suppose to the contrary that there exist $y \in Y'_1$, $b \in N_G(y) - Y'_1$ and $f \in F \cap L'(b) \cap L'(y)$. Since $f \notin L'(q)$ for every $q \in Y'_1 - Y_1$, we have $y \in Y_1$. Since $N^{\geq s}(Y_1) \subseteq W$ and $b \notin W$, we have $|N_G(b) \cap Y_1| \in [s - 1]$. That is, $b \in N^{< s}(Y_1)$. However, $f \in L'(b) \cap L'(y) \subseteq L(b) \cap L(y)$, contradicting that L satisfies (L3). Hence Statement 4 holds.

Now we prove Statement 5. Suppose there exists $a \in V(G) - Y'_1$ such that $L'(a) \cap F \neq L(a) \cap F$. If $a \notin N^{< s}(Y'_1)$, then $L'(a) = L(a)$, a contradiction. Thus $a \in N^{< s}(Y'_1)$. Since $Y'_1 \supseteq Y_1$, either $a \in N^{< s}(Y_1)$ or $a \in V(G) - N_G[Y_1]$. In either case, $|L(a)| = s + r - |N_G(a) \cap Y_1|$ by (L3) and (L4). Hence

$$\begin{aligned} |L'(a)| &= |L(a)| - |N_G(a) \cap (W - Y_1)| = s + r - |N_G(a) \cap Y_1| - |N_G(a) \cap (W - Y_1)| \\ &= s + r - |N_G(a) \cap Y'_1| \\ &\geq r + 1 \\ &> |F| \\ &\geq |L(a) \cap F|. \end{aligned}$$

Since $L'(a)$ is chosen so that $|L'(a) \cap F|$ is maximum among all subsets of $L(v) - \bigcup_{w \in N_G(v) \cap (W - Y_1)} L'(w)$ of size $|L(v)| - |N_G(v) \cap (W - Y_1)|$, and $\{y \in W - Y_1 : L'(y) \cap F \neq \emptyset\} = \emptyset$, we know $L'(a)$ must contain $L(a) \cap F$. Since $L'(a) \subseteq L(a)$, we have $L'(a) \cap F \subseteq L(a) \cap F$. Hence $L'(a) \cap F = L(a) \cap F$, a contradiction. So Statement 5 holds.

To complete the proof it suffices to show that L' is an (s, r, Y'_1) -list-assignment of G .

We first show that L' satisfies (L3). Let $v \in N^{< s}(Y'_1)$. Since $Y_1 \subseteq Y'_1$, either $N_G(v) \cap Y_1 = \emptyset$ or $v \in N^{< s}(Y_1)$. If $N_G(v) \cap Y_1 = \emptyset$, then

$$|L'(v)| = |L(v)| - |N_G(v) \cap (W - Y_1)| = s + r - |N_G(v) \cap Y'_1| \geq r + 1,$$

and $L'(v)$ is a subset of $L(v) - \{L'(w) : w \in N_G(v) \cap Y'_1\}$, so $L'(v) \cap L'(u) = \emptyset$ for every $u \in N_G(v) \cap Y'_1$. So we may assume $v \in N^{<s}(Y_1)$. Since L satisfies (L3), we have $|L(v)| = s + r - |N_G(v) \cap Y_1| \geq 2$ and $L(v) \cap L(u) = \emptyset$ for every $u \in N_G(v) \cap Y_1$. Hence

$$\begin{aligned} |L'(v)| &= |L(v)| - |N_G(v) \cap (W - Y_1)| \\ &= s + r - |N_G(v) \cap Y_1| - |N_G(v) \cap (W - Y_1)| \\ &= s + r - |N_G(v) \cap (W \cup Y_1)| \\ &= s + r - |N_G(v) \cap Y'_1|. \end{aligned}$$

Furthermore, $L'(v) \subseteq L(v) - \{L'(w) : w \in W - Y_1\}$ and $L'(u) = L(u)$ for every $u \in Y_1$, so $L'(v) \cap L'(u) = \emptyset$ for every $u \in N_G(v) \cap (W \cup Y_1) = N_G(v) \cap Y'_1$. Hence L' satisfies (L3).

Let x be a vertex in $V(G) - N_G[Y'_1]$. Since $Y_1 \subseteq Y'_1$, we have $x \in V(G) - N_G[Y_1]$. Since L satisfies (L4), we have $|L(x)| = s + r$. Since $Y'_1 \cup N_G(Y'_1) \supseteq Y'_1 \cup N^{<s}(Y'_1)$, we have $L'(x) = L(x)$ which has size $s + r$. This shows that L' satisfies (L4).

Let $z \in N^{\geq s}(Y'_1)$. Then $|L'(z)| = |L(z)|$. Since $z \notin Y_1$ and L satisfies (L2)–(L5), we have $|L'(z)| \geq r + 1$. Since L' satisfies (L3) and (L4), it implies that L' satisfies (L5). Since L' satisfies (L3)–(L5), L' satisfies (L1) and (L2). Therefore L' is an (s, r, Y'_1) -list-assignment. $\square$

Let G be a graph, $\eta \in \mathbb{N}$, and g a nondecreasing function. Let L be a list-assignment of G , and let $Y_1 := \{v \in V(G) : |L(v)| = 1\}$. Then an L -coloring c is (η, g) -bounded if:

- the union of the monochromatic components with respect to c intersecting Y_1 contains at most $|Y_1|^2 g(|Y_1|)$ vertices, and
- every monochromatic component with respect to c contains at most $\eta^2 g(\eta)$ vertices.

Lemma 13. *For all $s, t, k \in \mathbb{N}$, there exist a number $\eta > k$ and a nondecreasing function g with domain $\mathbb{N}_0$ and with $g(0) \geq \eta$ such that if G is a graph with no $K_{s,t}$ subgraph, $r \in \mathbb{N}$, Y_1 is a subset of $V(G)$ with $|Y_1| \leq \eta$, F is a set of colors with $|F| \leq r - 1$, and L is an (s, r, Y_1) -list-assignment of G such that $\{y \in Y_1 : x \in L(y)\}$ is a stable set in G for every $x \in F$, then one of the following holds:*

1. *There exists an (η, g) -bounded L -coloring of G such that for every $x \in F$, the set of vertices colored x is a stable set in G .*
2. $|Y_1| > k$.
3. *For every color ℓ , there exist a subset Y'_1 of $V(G)$ with $\eta \geq |Y'_1| > |Y_1|$ and an (s, r, Y'_1) -list-assignment L' of G with $L'(v) \subseteq L(v)$ for every $v \in V(G)$, such that:*

- (a) *there does not exist an (η, g) -bounded L' -coloring of G such that for every $x \in F$, the set of vertices colored x is a stable set in G ,*

- (b) for every L' -coloring of G , every monochromatic component intersecting Y_1 is contained in $G[Y_1]$,
- (c) $\{v \in Y_1 : F \cap L(v) \neq \emptyset\} = \{v \in Y'_1 : F \cap L'(v) \neq \emptyset\}$,
- (d) for every $x \in F \cup \{\ell\}$ and $y \in Y'_1$ with $x \in L'(y)$, we have $\{v \in N_G(y) - Y'_1 : x \in L'(v)\} = \emptyset$, and
- (e) for every $v \in V(G) - Y'_1$, we have $L'(v) \cap F = L(v) \cap F$.

4. $Y_1 \neq \emptyset$, $N_G(Y_1) = \emptyset$, and there does not exist an (η, g) -bounded $L|_{G-Y_1}$ -coloring of $G - Y_1$ such that for every $x \in F$, the set of vertices colored x is a stable set in $G - Y_1$.

Proof. Let f be the function $f_{s,t}$ in Lemma 11. Let $h_{-1} : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ be the identity function, and let $h_0 : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ be the function defined by $h_0(x) := x + f(x)$ for $x \in \mathbb{N}_0$. For $i \geq 1$, let $h_i : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ be the function defined by $h_i(x) := h_0(h_{i-1}(x))$ for $x \in \mathbb{N}_0$. Let $\eta := \max\{h_k(k), k + 1\}$. Let $g : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ to be the function defined by $g(x) := h_x(x) + \eta$ for $x \in \mathbb{N}_0$.

Suppose that Statements 1, 2 and 4 do not hold. Suppose to the contrary that Statement 3 does not hold for some color ℓ .

First suppose that $Y_1 = \emptyset$. Define R to be a $(\{v\}, F \cup \{\ell\})$ -progress of L , where v is a vertex of G . If G has an (η, g) -bounded R -coloring such that for every $x \in F$, the set of vertices colored x is a stable set, then G has an (η, g) -bounded L -coloring such that for every $x \in F$, the set of vertices colored x is a stable set, so Statement 1 holds, a contradiction. So G has no (η, g) -bounded R -coloring such that for every $x \in F$, the set of vertices colored x is a stable set, then Statement 3 holds if we take $Y'_1 = \{v\}$ and $L' = R$ by Lemma 12.

Hence $Y_1 \neq \emptyset$. Suppose that $N_G(Y_1) = \emptyset$. If $V(G) = Y_1$, then there exists an (η, g) -bounded L -coloring of G such that for every $x \in F$, the set of vertices colored x is a stable set, a contradiction. So $V(G) \neq Y_1$. Define $G' = G - Y_1$. Since Statement 4 does not hold, there exists an (η, g) -bounded $L|_{G'}$ -coloring of G' such that for every $x \in F$, the set of vertices colored x is a stable set. We can further color the vertices in Y_1 by the unique element in their lists to obtain an (η, g) -bounded L -coloring of G such that for every $x \in F$, the set of vertices colored x is a stable set, a contradiction.

Hence $Y_1 \neq \emptyset$ and $N_G(Y_1) \neq \emptyset$. Let $Y_2 = N^{<s}(Y_1)$.

Suppose that $N^{\geq s}(Y_1) = \emptyset$. So $Y_2 = N_G(Y_1)$. Since $N_G(Y_1) \neq \emptyset$, there exists a vertex $z \in Y_2$. Define $Y'_1 := Y_1 \cup \{z\}$ and define L' to be a $(\{z\}, F \cup \{\ell\})$ -progress of L . Since Statement 2 does not hold, $|Y'_1| = |Y_1| + 1 \leq k + 1 \leq \eta$. Since $Y_2 = N_G(Y_1)$ and by (L3) $L(v) \cap L(u) = \emptyset$ for every $v \in Y_2$ and $u \in N_G(v) \cap Y_1$, the union of the monochromatic components with respect to any L' -coloring of G intersecting Y_1 is contained in $G[Y_1]$. Since Statement 3 does not hold, by Lemma 12, there exists an (η, g) -bounded L' -coloring c' of G such that for every $x \in F$, the set of vertices colored x is a stable set. In particular, the union of the monochromatic components with respect to c' intersecting Y_1 is contained in $G[Y_1]$. So c' is an (η, g) -bounded L -coloring of G , and Statement 1 holds, a contradiction.

Hence $N^{\geq s}(Y_1) \neq \emptyset$. Denote Y_1 by $\{y_1, y_2, \dots, y_{|Y_1|}\}$. For every $i \in [|Y_1|]$, let ℓ_i be the unique element of $L(y_i)$; let $\ell_{|Y_1|+1} = \ell$. Define $L_0 = L$ and $U_0 = Y_1$. For $i \geq 1$, let L_i be a $(N^{\geq s}(U_{i-1}), \{\ell_i\} \cup F)$ -progress and define $U_i := \{v \in V(G) : |L_i(v)| = 1\}$. Define $L^* := L_{|Y_1|+1}$ and $Y_1^* := U_{|Y_1|+1}$.

Claim 13.1. $|Y_1^*| \leq h_{|Y_1|}(|Y_1|)$.

Proof. We shall prove that $|U_i| \leq h_{i-1}(|Y_1|)$ by induction on $i \geq 0$. When $i = 0$, we have $|U_0| = |Y_1| = h_{-1}(|Y_1|)$. Now assume that $i \geq 1$ and the claim holds for all smaller i . By induction and Lemma 11,

$$\begin{aligned} |U_i| &= |U_{i-1}| + |N^{\geq s}(U_{i-1})| \leq |U_{i-1}| + f(|U_{i-1}|) = h_0(|U_{i-1}|) \leq h_0(h_{i-2}(|Y_1|)) \\ &= h_{i-1}(|Y_1|). \end{aligned}$$

The case $i = |Y_1| + 1$ proves the claim. $\square$

Since Statement 2 does not hold, $|Y_1| \leq k$. By Claim 13.1, $|Y_1^*| \leq h_k(k) \leq \eta$. Recall that we proved that $N^{\geq s}(U_0) \neq \emptyset$. So $|Y_1^*| > |Y_1|$. And by Lemma 12, $\{v \in Y_1 : F \cap L(v) \neq \emptyset\} = \{v \in Y_1^* : F \cap L^*(v) \neq \emptyset\}$.

Claim 13.2. For every L^* -coloring c of G , every monochromatic component with respect to c intersecting Y_1 is contained in Y_1^* .

Proof. For $i \in [|Y_1|]$, let M_i be the monochromatic component with respect to c containing y_i . We shall prove that $V(M_i) \subseteq U_{i-1}$ for every $i \in [|Y_1|]$.

For $i \in [|Y_1|]$, note that $U_i - U_{i-1} = N^{\geq s}(U_{i-1})$. Since L_i is an $(U_i - U_{i-1}, \{\ell_i\} \cup F)$ -progress, $\ell_i \notin L_i(u)$ for every $u \in U_i - U_{i-1}$, and for every $v \in N^{< s}(U_{i-1})$, either $v \notin N_G(V(M_i) \cap U_{i-1})$ or $\ell_i \notin L_i(v)$. Since $L_i(v) \supseteq L^*(v)$ for every $v \in V(G)$, we have $V(M_i) \cap N_G(U_{i-1}) = \emptyset$ since M_i is connected. Since M_i is connected and $V(M_i) \cap U_{i-1} \supseteq V(M_i) \cap U_0 \neq \emptyset$, we have $V(M_i) \subseteq U_{i-1}$.

Every monochromatic component M with respect to c intersecting Y_1 equals M_j for some $j \in [|Y_1|]$, so $V(M) \subseteq U_{j-1} \subseteq Y_1^*$. $\square$

Since L^* is an $(N^{\geq s}(U_{|Y_1|}), \{\ell\} \cup F)$ -progress of $L_{|Y_1|}$, by Lemma 12, for every $x \in F \cup \{\ell\}$ and $y \in Y_1^*$ with $x \in L^*(y)$, we have $\{v \in N_G(y) - Y_1^* : x \in L^*(v)\} = \emptyset$. Since Statement 3 does not hold, by Claim 13.2, there exists an (η, g) -bounded L^* -coloring c^* of G such that for every $x \in F$, the set of vertices colored x is a stable set. So every monochromatic component with respect to c^* contains at most $\eta^2 g(\eta)$ vertices. By Claims 13.1 and 13.2, the union of the monochromatic components with respect to c^* intersecting Y_1 contains at most $|Y_1^*| \leq h_{|Y_1|}(|Y_1|) \leq g(|Y_1|)$ vertices. Therefore, c^* is an (η, g) -bounded L -coloring of G such that for every $x \in F$, the set of vertices colored x is a stable set. So Statement 1 holds. This contradiction proves the lemma. $\square$

2.1. Layered treewidth

We now take an excursion to introduce a tool used in our main proofs. A *layering* of graph G is a partition $(V_1, \dots, V_n)$ of $V(G)$ such that for each edge $vw \in E(G)$ there exists $i \in [n-1]$ such that $v, w \in V_i \cup V_{i+1}$. The *layered treewidth* of a graph G is the minimum integer ℓ such that G has a tree decomposition $(T, \mathcal{X} = (X_x : x \in V(T)))$ and a layering $(V_1, \dots, V_n)$, such that $|X_x \cap V_i| \leq \ell$ for each $x \in V(T)$ and $i \in [n]$.

Layered treewidth was introduced by Dujmović et al. [11]. They proved that every planar graph has layered treewidth at most 3; more generally, that every graph with Euler genus at most g has layered treewidth at most $2g + 3$; and most generally, that a minor-closed class has bounded layered treewidth if and only if it excludes some apex graph as a minor. Several interesting non-minor-closed classes also have bounded layered treewidth [2,9,10].

In our companion paper [33], we prove that graphs of bounded layered treewidth and with no $K_{s,t}$ subgraph are $(s+2)$ -colorable with bounded clustering. In fact, we prove the following stronger result.

Theorem 14 ([33]). *For all $s, t, w, \xi \in \mathbb{N}$ there exists $\eta \in \mathbb{N}$ such that if G is a graph with no $K_{s,t}$ subgraph such that $G - Z$ has layered treewidth at most w for some $Z \subseteq V(G)$ with $|Z| \leq \xi$, then G is $(s+2)$ -colorable with clustering η .*

We actually need the following more precise result that fits our list coloring setup. Let G be a graph and $Z \subseteq V(G)$. A Z -*layering* $\mathcal{V}$ of G is an ordered partition $(V_1, V_2, \dots)$ of $V(G) - Z$ such that for every edge e of $G - Z$, there exists $i \in \mathbb{N}$ such that both endpoints of e are contained in $V_i \cup V_{i+1}$. For a tree decomposition $(T, \mathcal{X})$ of G , the $\mathcal{V}$ -*width* is

$$\max_{i \in \mathbb{N}} \max_{t \in V(T)} |X_t \cap V_i|.$$

Let $s \in \mathbb{N}$. A list-assignment L of G is $(s, \mathcal{V})$ -*compatible* if:

- $L(v) \subseteq [s+2]$ for every $v \in V(G)$, and
- $i \notin L(v)$ for every $i \in [s+2]$ and $v \in \bigcup (V_j : j \equiv i \pmod{s+2})$.

For $Y_1 \subseteq V(G)$, we say that (Y_1, L) is a $\mathcal{V}$ -*standard pair* if L is an $(s, 1, Y_1)$ -list-assignment and is $(s, \mathcal{V})$ -compatible.

Theorem 15 ([33]). *For all $s, t, w, k, \xi \in \mathbb{N}$, there exists $\eta^* \in \mathbb{N}$ such that if G is a graph with no $K_{s,t}$ subgraph, Z is a subset of $V(G)$ with $|Z| \leq \xi$, $\mathcal{V}$ is a Z -layering of G , $(T, \mathcal{X})$ is a tree decomposition of $G - Z$ with $\mathcal{V}$ -width at most w , Y_1 is a subset of $V(G)$ with $|Y_1| \leq k$, L is an $(s, \mathcal{V})$ -compatible list-assignment of G such that (Y_1, L) is a $\mathcal{V}$ -standard pair, then there exists an L -coloring of G with clustering η^* .*

3. Tangles and treewidth

The goal of this section is to prove Theorem 5 regarding graphs of bounded treewidth and to set-up machinery for the proofs of Theorems 2 and 7 in subsequent sections.

Lemma 16. *For all $s, t, \theta, \eta, r \in \mathbb{N}$ with $\eta \geq 9\theta + 1$, for every nondecreasing function g with domain $\mathbb{N}_0$, if G is a graph with no $K_{s,t}$ subgraph, Y_1 is a subset of $V(G)$ with $9\theta + 1 \leq |Y_1| \leq \eta$, F is a set of colors with $|F| \leq r$, and L is an (s, r, Y_1) -list-assignment of G such that $\{y \in Y_1 : x \in L(y)\}$ is a stable set in G for every $x \in F$, then at least one of the following holds:*

1. *There exists an (η, g) -bounded L -coloring of G such that for every $x \in F$, the set of vertices colored x is a stable set in G .*
2. *There exist an induced subgraph G' of G with $|V(G')| < |V(G)|$, a subset Y'_1 of $V(G')$ with $|Y'_1| \leq \eta$, and an (s, r, Y'_1) -list-assignment L' of G' such that the following hold:*
 - (a) $L'(v) \subseteq L(v)$ for every $v \in V(G')$.
 - (b) *There does not exist an (η, g) -bounded L' -coloring of G' such that for every $x \in F$, the set of vertices colored x is a stable set in G' .*
 - (c) $\{v \in Y'_1 : F \cap L(v) \neq \emptyset\} \cap V(G') = \{v \in Y'_1 : F \cap L'(v) \neq \emptyset\}$.
 - (d) *For every $v \in V(G') - Y'_1$, we have $L'(v) \cap F = L(v) \cap F$.*
3. $\mathcal{T} := \{(A, B) : |V(A \cap B)| < \theta, |V(A) \cap Y_1| \leq 3\theta\}$ *is a tangle of order θ in G .*

Proof. Suppose that Statements 1, 2 and 3 do not hold. Since $\mathcal{T}$ is not a tangle, one of (T1), (T2) or (T3) is violated.

Suppose that (T2) violated. So there exist $(A_i, B_i) \in \mathcal{T}$ for $i \in [3]$ such that $A_1 \cup A_2 \cup A_3 = G$. Hence $|Y_1| \leq \sum_{i=1}^3 |A_i \cap Y_1| \leq 9\theta$, a contradiction. So $\mathcal{T}$ satisfies (T2). Similarly, for every $(A, B) \in \mathcal{T}$, we have $V(A) \neq V(G)$; otherwise $|Y_1| = |V(A) \cap Y_1| \leq 3\theta$, a contradiction. So $\mathcal{T}$ satisfies (T3).

Therefore, (T1) is violated. So there exists a separation (A, B) of G of order less than θ such that $(A, B) \notin \mathcal{T}$ and $(B, A) \notin \mathcal{T}$. That is, $|V(A) \cap Y_1| > 3\theta$ and $|V(B) \cap Y_1| > 3\theta$. In particular, $V(A) \neq V(G)$ and $V(B) \neq V(G)$.

Let $Y_A := (Y_1 \cap V(A)) \cup V(A \cap B)$ and $Y_B := (Y_1 \cap V(B)) \cup V(A \cap B)$. Note that $\max\{|Y_A|, |Y_B|\} \leq |Y_1| - 2\theta < \eta$. Let $G_A := G[V(A)]$ and $G_B := G[V(B)]$.

Let L' be a $(V(A \cap B), F)$ -progress L' of L . Since $|F| \leq r$ and L is an (s, r, Y_1) -list-assignment of G , by Lemma 12, L' is an $(s, r, Y_A \cup Y_B)$ -list-assignment of G . Let $L_A = L'|_{V(A)}$ and $L_B = L'|_{V(B)}$. Since $V(A \cap B) \subseteq Y_A \cap Y_B$, L_A is an (s, r, Y_A) -list-assignment of G_A , and L_B is an (s, r, Y_B) -list-assignment of G_B . Moreover, by Lemma 12, for every $C \in \{A, B\}$, L_C is an (s, r, Y_C) -list-assignment of G_A such that

- $L_C(v) \subseteq L(v)$ for every $v \in V(G_C)$.

- $\{v \in Y_1 : F \cap L(v) \neq \emptyset\} \cap V(C) = \{v \in Y_C : F \cap L_C(v) \neq \emptyset\}$.
- For every $v \in V(G_C) - Y_C$, we have $L_C(v) \cap F = L(v) \cap F$.

Since Statement 2 does not hold, for $C \in \{A, B\}$, there exists an (η, g) -bounded L_C -coloring c_C of G_C such that for every $x \in F$, the set of vertices colored x is a stable set in G_C . By construction, $c_A(v) = c_B(v)$ for every $v \in V(A \cap B)$. Define $c(v) := c_A(v)$ if $v \in V(A)$ and define $c(v) := c_B(v)$ if $v \in V(B)$. Clearly, c is an L -coloring such that for every $x \in F$, the set of vertices colored x is a stable set in G .

Let C be the union of the monochromatic components with respect to c intersecting $Y_1 \cup V(A \cap B)$. Then $V(C) \cap Y_A \neq \emptyset$ and $V(C) \cap Y_B \neq \emptyset$. By construction,

$$\begin{aligned} |V(C)| &\leq |V(C) \cap V(A)| + |V(C) \cap V(B)| \\ &\leq |Y_A|^2 g(|Y_A|) + |Y_B|^2 g(|Y_B|) \\ &\leq (|Y_A|^2 + |Y_B|^2) g(|Y_1|) \\ &\leq ((|Y_A| + |Y_B|)^2 - 2|Y_A||Y_B|) g(|Y_1|). \end{aligned}$$

Since $|Y_A| \geq 3\theta$ and $|Y_B| \geq 3\theta$,

$$|Y_1|\theta + 2\theta^2 \leq (|Y_A| + |Y_B|)\theta + 2\theta^2 \leq |Y_A| \cdot \frac{|Y_B|}{3} + |Y_B| \cdot \frac{|Y_A|}{3} + 2 \cdot \frac{|Y_A|}{3} \cdot \frac{|Y_B|}{3} \leq |Y_A| \cdot |Y_B|.$$

Therefore,

$$\begin{aligned} |V(C)| &\leq ((|Y_A| + |Y_B|)^2 - 2|Y_A||Y_B|) g(|Y_1|) \\ &\leq ((|Y_1| + \theta)^2 - 2|Y_1|\theta - 4\theta^2) g(|Y_1|) \\ &\leq |Y_1|^2 g(|Y_1|). \end{aligned}$$

Since Statement 1 does not hold, c is not an (η, g) -bounded L -coloring of G . Since $|Y_1|^2 g(|Y_1|) \leq \eta^2 g(\eta)$, there exists a monochromatic component M with respect to c disjoint from $Y_1 \cup V(A \cap B)$ containing at least $\eta^2 g(\eta) + 1$ vertices. However, since M is disjoint from $V(A \cap B)$, either M is contained in G_A or M is contained in G_B . So M is a monochromatic component with respect to c_A or c_B . Since c_A, c_B are (η, g) -bounded, M contains at most $\eta^2 g(\eta)$ vertices, a contradiction. This proves the lemma. $\square$

Theorem 17. For all $s, t, w \in \mathbb{N}$, there exist $\eta \in \mathbb{N}$ and a nondecreasing function g such that if G is a graph of treewidth at most w and with no $K_{s,t}$ subgraph, Y_1 is a subset of $V(G)$ with $|Y_1| \leq \eta$, and L is an $(s, 1, Y_1)$ -list-assignment of G , then there exists an (η, g) -bounded L -coloring of G .

Proof. Define η and g to be the number η and the function g in Lemma 13 by taking $s = s, t = t, k = 9w + 18$ and $r = 1$. Note that $g(x) \geq \eta > 9w + 18$ for every $x \in \mathbb{N}_0$ by Lemma 13.

Suppose to the contrary that this theorem does not hold. So there exist a graph of treewidth at most w and with no $K_{s,t}$ subgraph, a subset Y_1 of $V(G)$ with $|Y_1| \leq \eta$, and an $(s, 1, Y_1)$ -list-assignment L of G such that there does not exist an (η, g) -bounded L -coloring of G . We further assume that $|V(G)|$ is as small as possible and subject to this, $|Y_1|$ is as large as possible. Since $g(x) \geq \eta$ for every $x \in \mathbb{N}_0$, we have $|V(G)| > \eta$, as otherwise any L -coloring of G is (η, g) -bounded.

By Lemma 13 and the choice of G and Y_1 , we have $|Y_1| > 9w + 18$. By Lemma 16, there exists a tangle of order $w + 2$ in G . But G has treewidth at most w , there exists no tangle of order $w + 2$ in G by [48, Lemma (5.2)], a contradiction. This proves the theorem. $\square$

Corollary 18. *For all $s, t, w \in \mathbb{N}$, there exists $\eta \in \mathbb{N}$ and a nondecreasing function g with domain $\mathbb{N}_0$ such that if G is a graph of treewidth at most w and with no $K_{s,t}$ subgraph and L is a $(s + 1)$ -list-assignment of G , then there exists an L -coloring of G with clustering $\eta^2 g(\eta)$.*

Proof. Define $L'(v)$ to be an $(s + 1)$ -element subset of $L(v)$ for every $v \in V(G)$. Clearly, L' is an $(s, 1, \emptyset)$ -list-assignment of G , and every L' -coloring is an L -coloring. The result immediately follows from Theorem 17. $\square$

Observe that Corollary 18 implies Theorem 5.

4. Graph minor structure theorem

Our proofs of Theorems 2 and 7 depend on the Graph Minor Structure Theorem, which we now introduce. Recall the definition of an H -minor α in a graph G from Section 1.8. A tangle $\mathcal{T}$ in G controls an H -minor α if there does not exist $(A, B) \in \mathcal{T}$ of order less than $|V(H)|$ such that $V(\alpha(h)) \subseteq V(A)$ for some $h \in V(H)$. We use the following theorem of Geelen et al. [18] on odd minors.

Theorem 19 ([18, Theorem 13]). *There is a constant c such that for all $\ell \in \mathbb{N}$, if $t := \lceil c\ell\sqrt{\log 12\ell} \rceil$ then for every graph G that contains a K_t -minor α , either G contains an odd K_ℓ -minor, or there exists a set X of vertices with $|X| < 8\ell$ such that the (unique) block U of $G - X$ that intersects all branch sets of α disjoint from X is bipartite.*

A *society* is a pair (S, Ω) , where S is a graph and Ω is a cyclic permutation of a subset $\overline{\Omega}$ of $V(S)$. For $\rho \in \mathbb{N}_0$, a society (S, Ω) is a ρ -vortex if for all distinct $u, v \in \overline{\Omega}$, there do not exist $\rho + 1$ disjoint paths in S between $I \cup \{u\}$ and $J \cup \{v\}$, where I is the set of vertices in $\overline{\Omega}$ after u and before v in the order Ω , and J is the set of vertices in $\overline{\Omega}$ after v and before u . For a society (S, Ω) with $\overline{\Omega} = (v_1, v_2, \dots, v_n)$ in order, a *vortical decomposition* of (S, Ω) is a path decomposition $(t_1 t_2 \dots t_n, \mathcal{X})$ such that the i -th bag X_i of $\mathcal{X}$ contains the i -th vertex v_i for each $i \in [n]$. The *adhesion* of such a vortical

decomposition is $\max\{|X_i \cap X_j| : i, j \in [n], i \neq j\}$. We use the following theorem of Robertson and Seymour [47].

Theorem 20 ([47, (8.1)]). *Every ρ -vortex has a vortical decomposition of adhesion at most ρ .*

A *segregation* of a graph G is a set $\mathcal{S}$ of societies such that:

- S is a subgraph of G for every $(S, \Omega) \in \mathcal{S}$, and $\bigcup\{S : (S, \Omega) \in \mathcal{S}\} = G$, and
- for all distinct (S, Ω) and $(S', \Omega') \in \mathcal{S}$, we have $V(S \cap S') \subseteq \overline{\Omega} \cap \overline{\Omega'}$ and $E(S \cap S') = \emptyset$.

We write $V(\mathcal{S}) = \bigcup\{\overline{\Omega} : (S, \Omega) \in \mathcal{S}\}$. For positive integers κ, ρ , a segregation $\mathcal{S}$ is of *type* (κ, ρ) if there exist disjoint subsets $\mathcal{S}_1, \mathcal{S}_2$ of $\mathcal{S}$ with $\mathcal{S} = \mathcal{S}_1 \cup \mathcal{S}_2$ and $|\mathcal{S}_2| \leq \kappa$ such that $|\overline{\Omega}| \leq 3$ for every $(S, \Omega) \in \mathcal{S}_1$, and every member of $\mathcal{S}_2$ is a ρ -vortex. For a tangle $\mathcal{T}$ in G , a segregation $\mathcal{S}$ of G is $\mathcal{T}$ -*central* if for every $(S, \Omega) \in \mathcal{S}$, there exists no $(A, B) \in \mathcal{T}$ with $B \subseteq S$.

Let Σ be a surface. For every subset Δ of Σ , we denote the closure of Δ by $\overline{\Delta}$ and the boundary of Δ by $\partial\Delta$. An *arrangement* of a segregation $\mathcal{S} = \{(S_1, \Omega_1), \dots, (S_k, \Omega_k)\}$ in Σ is a function α with domain $\mathcal{S} \cup V(\mathcal{S})$, such that:

- For $i \in [k]$, $\alpha(S_i, \Omega_i)$ is a closed disk $\Delta_i \subseteq \Sigma$, and $\alpha(x) \in \partial\Delta_i$ for each $x \in \overline{\Omega_i}$.
- For $i, j \in [k]$ with $i \neq j$, if $x \in \Delta_i \cap \Delta_j$, then $x = \alpha(v)$ for some $v \in \overline{\Omega_i} \cap \overline{\Omega_j}$.
- For all distinct $x, y \in V(\mathcal{S})$, we have $\alpha(x) \neq \alpha(y)$.
- For $i \in [k]$, Ω_i is mapped by α to a natural order of $\alpha(\overline{\Omega_i})$ determined by $\partial\Delta_i$.

An arrangement is *proper* if $\Delta_i \cap \Delta_j = \emptyset$ whenever $|\overline{\Omega_i}|, |\overline{\Omega_j}| > 3$, for all $1 \leq i < j \leq k$.

For a tangle $\mathcal{T}$ in a graph G of order θ and a subset Z of $V(G)$ with $|Z| < \theta$, $\mathcal{T} - Z$ is defined to be the set of all separations (A', B') of $G - Z$ of order less than $\theta - |Z|$ such that there exists $(A, B) \in \mathcal{T}$ with $Z \subseteq V(A \cap B)$, $A' = A - Z$ and $B' = B - Z$. It is proved in Robertson and Seymour [48] that $\mathcal{T} - Z$ is a tangle in $G - Z$ of order $\theta - |Z|$.

The following is the Graph Minor Structure Theorem of Robertson and Seymour [49].

Theorem 21 ([49, (3.1)]). *For every graph H , there exist $\kappa, \rho, \xi, \theta \in \mathbb{N}$ such that if $\mathcal{T}$ is a tangle of order at least θ in a graph G controlling no H -minor of G , then there exist $Z \subseteq V(G)$ with $|Z| \leq \xi$ and a $(\mathcal{T} - Z)$ -central segregation $\mathcal{S}$ of $G - Z$ of type (κ, ρ) such that $\mathcal{S}$ has a proper arrangement in some surface in which H cannot be embedded.*

5. Main proofs

The goal of this section is to prove Theorems 2 and 7. Let G be a graph. A *location* $\mathcal{L}$ in G is a collection of separations of G such that $A \subseteq B'$ for every ordered pair of distinct members $(A, B), (A', B')$ of $\mathcal{L}$. Define $G(\mathcal{L})$ to be the graph $G[\bigcap_{(A, B) \in \mathcal{L}} V(B)]$.

Let $s \in \mathbb{N}$, $r \in \mathbb{N}_0$ and $\ell \in [0, s + 2]$. Let G be a graph and let $Y_1 \subseteq V(G)$. A list-assignment L of G is an (s, Y_1, ℓ, r) -list-assignment if the following hold:

- (R1) $L(v) \subseteq [s + 2 + r]$ for all $v \in V(G)$.
- (R2) L is an $(s, r + 2, Y_1)$ -list-assignment of G .
- (R3) For every $y \in Y_1$ and color $x \in \{\ell\} \cup [s + 3, s + 2 + r]$ with $x \in L(y)$, we have $\{v \in N_G(y) - Y_1 : x \in L(v)\} = \emptyset$.
- (R4) For every $x \in [s + 3, s + 2 + r]$, the set $\{y \in Y_1 : x \in L(y)\}$ is a stable set in G .
- (R5) For every $v \in V(G) - Y_1$,

$$|\{y \in N_G(v) \cap Y_1 : L(y) \subseteq [s + 3, s + r + 2]\}| = r - |L(v) \cap [s + 3, s + 2 + r]|.$$

Note that every $(s, Y_1, 0, 0)$ -list-assignment of G is an $(s, 2, Y_1)$ -list-assignment. For graph minors we use $\ell = r = 0$. The more general setting is used for odd minors.

Let G be a graph and $Z, Y_1 \subseteq V(G)$. Let $s \in \mathbb{N}$, $r \in \mathbb{N}_0$ and $\ell \in [0, s + 2]$. Let L be an (s, Y_1, ℓ, r) -list-assignment of G . A (Z, ℓ) -growth of L is a list-assignment L' of G defined as follows:

- Let $Y_1^{(-1)} := Y_1$ and $L^{(-1)}(v) := L(v)$ for every $v \in V(G)$.
- Let $U_0 := Z$, and for each $i \geq 1$, let $U_i := N^{\geq s}(Y_1^{(i-1)})$.
- Let $Y_1^{(i)} := Y_1^{(i-1)} \cup U_i$ for $i \geq 0$.
- For every $i \geq 0$, let $L^{(i)}$ be a $(U_i, \{\ell, i\} \cup [s + 3, s + 2 + r])$ -progress of $L^{(i-1)}$ such that $L^{(i)}(v) \cap [s + 3, s + 2 + r] = L^{(i-1)}(v) \cap [s + 3, s + 2 + r]$ for every $v \in V(G) - (Y_1^{(i-1)} \cup U_i)$. (Note that $|\{\ell, i\} \cup [s + 3, s + 2 + r]| \leq r + 2$, so a $(U_i, \{\ell, i\} \cup [s + 3, s + 2 + r])$ -progress of $L^{(i-1)}$ can be defined, by (R2); furthermore, such an $L^{(i)}$ exists since $s + 2 \geq s - 1$ and for every $v \in V(G) - (Y_1^{(i-1)} \cup U_i)$, $L^{(i)}(v)$ can be obtained from $L^{(i-1)}(v)$ by removing elements, and $L^{(i)}(v) \neq L^{(i-1)}(v)$ only when $v \in N^{< s}(Y_1^{(i-1)} \cup U_i)$.)
- Let $Y_1' := Y_1^{(s+2)}$ and $L' := L^{(s+2)}$.

Lemma 22. For all $s, t \in \mathbb{N}$ there exists a function $h : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ such that if G is a graph with no $K_{s,t}$ subgraph, $Z, Y_1 \subseteq V(G)$, $\ell \in [0, s + 2]$, $r \in \mathbb{N}_0$, and L is an (s, Y_1, ℓ, r) -list-assignment of G , then for each (Z, ℓ) -growth L' of L and set Y_1' defined above the following hold:

1. L' is an (s, Y_1', ℓ, r) -list-assignment.
2. $L'(v) \subseteq L(v)$ for every $v \in V(G)$.
3. $|Y_1'| \leq h(|Y_1 \cup Z|)$.
4. For every L' -coloring c' of G , every monochromatic component M with respect to c' intersecting $Y_1 \cup Z$ is contained in $G[Y_1']$. Furthermore, if the vertices of M are assigned colors in $\{\ell\} \cup [s + 3, s + 2 + r]$, then $M \subseteq G[Y_1]$.
5. If (A, B) is a separation of G with $Z \subseteq V(A \cap B)$, then $|Y_1' \cap V(A)| \leq h(|V(A \cap B)| + |Y_1 \cap V(A)|)$ for every $(A, B) \in \mathcal{L}$.

Proof. Let $f : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ be the function $f_{s,t}$ in Lemma 11. Let $f_0 : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ be the function defined by $f_0(x) := x + f(x)$ for $x \in \mathbb{N}_0$. For $i \geq 1$, let $f_i : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ be the function defined by $f_i(x) := x + f_{i-1}(x) + f(f_{i-1}(x))$. Define $h := f_{s+2}$.

Since L is an $(s, r+2, Y_1)$ -list-assignment and L' is obtained from L by repeatedly taking (W_i, F_i) -progress for some sets W_i, F_i with $|F_i| \leq r+2$ and $\{\ell\} \cup [s+3, s+2+r] \subseteq F_i$ for all i , such that $L'(v) \cap [s+3, s+2+r] = L(v) \cap [s+3, s+2+r]$ for every $v \in V(G) - Y'_1$, we know that L' is an $(s, r+2, Y'_1)$ -list-assignment of G satisfying (R5) such that

$$\begin{aligned} \{v \in Y'_1 : L'(v) \cap (\{\ell\} \cup [s+3, s+2+r]) \neq \emptyset\} \\ = \{v \in Y_1 : L(v) \cap (\{\ell\} \cup [s+3, s+2+r]) \neq \emptyset\} \end{aligned}$$

by Lemma 12. So L' satisfies (R1)–(R5), and Statement 2 holds. Hence Statements 1 and 2 hold.

For every $i \geq 0$, let $U_i, Y_1^{(i)}, L^{(i)}$ be the sets and list-assignment mentioned in the definition of a (Z, ℓ) -growth. By Lemma 11, it is easy to verify that $|U_i| \leq f(|Y_1^{(i-1)}|)$ and $|Y_1^{(i)}| \leq |Y_1^{(i-1)}| + |U_i| \leq f_i(|Y_1 \cup Z|)$ for every $i \geq 0$ by induction on i . This proves Statement 3.

Let c' be an L' -coloring of G . Let M_i be a monochromatic component with respect to c' intersecting $Y_1 \cup Z$ such that all vertices of M_i are colored i for some $i \in [s+r+2]$. If $i \in \{\ell\} \cup [s+3, s+2+r]$, then $V(M_i) \cap Z - Y_1 = \emptyset$ and $V(M_i) \subseteq Y_1$ since c' is an L -coloring and L satisfies (R3). So we may assume that $i \in [s+2] - \{\ell\}$. Since $L^{(i)}$ is a $(U_i, \{\ell, i\} \cup [s+3, s+2+r])$ -progress of $L^{(i-1)}$, we have $V(M_i) \cap U_i = \emptyset$. Since M_i is connected and $V(M_i) \cap U_i = \emptyset$ and $V(M_i) \cap (Y_1 \cup Z) \neq \emptyset$, we have either $V(M_i) \subseteq Y_1^{(i-1)}$ or there exists $xy \in E(M_i)$ such that $y \in Y_1^{(i-1)}$ and $x \in N_G(y) \cap N^{<s}(Y_1^{(i-1)})$. But the latter is impossible by (L3). Hence $V(M_i) \subseteq Y_1^{(i-1)} \subseteq Y_1^{(s+1)} \subseteq Y'_1$. Therefore, Statement 4 holds.

Let (A, B) be a separation of G with $Z \subseteq V(A \cap B)$. For every $i \geq 0$, since (A, B) is a separation,

$$U_i \cap V(A) \subseteq V(A \cap B) \cup \{v \in V(A) - Y_1^{(i-1)} : |N_G(v) \cap Y_1^{(i-1)} \cap V(A)| \geq s\}.$$

Since $Z \subseteq V(A \cap B)$, by Lemma 11, it is easy to prove by induction on i that $|U_i \cap V(A)| \leq |V(A \cap B)| + f(|Y_1^{(i-1)} \cap V(A)|)$ and $|Y_1^{(i)} \cap V(A)| \leq |Y_1^{(i-1)} \cap V(A)| + |U_i \cap V(A)| \leq f_i(|V(A \cap B)| + |Y_1 \cap V(A)|)$. Statement 5 holds by taking $i = s+2$. $\square$

Lemma 23. For all $s, t, t' \in \mathbb{N}$, there exist $\theta^* \in \mathbb{N}$ and nondecreasing functions g^*, η^* with domain $\mathbb{N}_0$ such that if G is a graph with no $K_{s,t}$ subgraph, $\theta \in \mathbb{N}$ with $\theta \geq \theta^*$, $\eta \in \mathbb{N}$ with $\eta \geq \eta^*(\theta)$, $Y_1 \subseteq V(G)$ with $3\theta < |Y_1| \leq \eta$, $\ell \in [0, s+2]$, $r \in \mathbb{N}_0$, L is an (s, Y_1, ℓ, r) -list-assignment of G , g is a nondecreasing function with domain $\mathbb{N}_0$ and with $g \geq g^*$, and $\mathcal{T}$ is a tangle in G of order θ that does not control a $K_{t'}$ -minor, where $\mathcal{T} = \{(A, B) : |V(A \cap B)| < \theta, |V(A) \cap Y_1| \leq 3\theta\}$, then either:

1. there exists an (η, g) -bounded L -coloring of G such that for every $x \in [s+3, s+2+r]$, the set of vertices colored x is a stable set, or
2. there exist $(A^*, B^*) \in \mathcal{T}$, a set Y_{A^*} with $|Y_{A^*}| \leq \eta^*(\theta)$ and $Y_1 \cap V(A^*) \subseteq Y_{A^*} \subseteq V(A^*)$, and an (s, Y_{A^*}, ℓ, r) -list-assignment L_{A^*} of $G[V(A^*)]$ such that there exists no (η, g) -bounded L_{A^*} -coloring of $G[V(A^*)]$ such that for every $x \in [s+3, s+2+r]$, the set of vertices colored x is a stable set.

Proof. Define the following:

- Let f be the function $f_{s,t}$ in Lemma 11.
- Let $f_0 : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ be the identity function. For every $i \in \mathbb{N}$, let $f_i : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ be the function defined by $f_i(x) := f_{i-1}(x) + f(f_{i-1}(x))$.
- Let $\kappa_0, \rho_0, \xi_0, \theta_0$ be the integers $\kappa, \rho, \xi, \theta$ in Theorem 21 taking $H = K_{t'}$.
- Let $h : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ be the function in Lemma 22 taking $s = s$ and $t = t$.
- Let $\theta^* := \theta_0 + 2\rho_0 + \xi_0 + 3$.
- Let $\eta^* : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ be the function defined by $\eta^*(x) := h(f_3(h(4x) + x))$ for every $x \in \mathbb{N}_0$.
- Let σ be the maximum Euler genus of a surface in which $K_{t'}$ cannot be embedded.
- For every $x \in \mathbb{N}_0$, let $w_0(x) := (3\kappa_0(2\sigma + 4)(2\rho_0 + 1) + 1) \cdot (f_1(h(4x) + x) + 1)$.
- For every $x \in \mathbb{N}_0$, let $\eta_1(x)$ be the number η^* in Theorem 15 taking $s = s$, $t = t + f_1(h(4x) + x) + (s + 2)^2$, $w = (s + 2)^2 \cdot w_0(x)$, $k = h(x + \xi_0) + s + 2$ and $\xi = h(x + \xi_0) + s + 2$.
- Let $g^* : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ be the function defined by $g^*(x) := \eta_1(2x) \cdot (1 + f_1(h(4x + \xi_0) + x + \xi_0) \cdot \eta^*(x))$ for every $x \in \mathbb{N}_0$.

Let G be a graph with no $K_{s,t}$ subgraph, $\theta \in \mathbb{N}$ with $\theta \geq \theta^*$, $\eta \in \mathbb{N}$ with $\eta \geq \eta^*(\theta)$, Y_1 a subset of $V(G)$ with $3\theta < |Y_1| \leq \eta$, $\ell \in [0, s+2]$, $r \in \mathbb{N}_0$, g a nondecreasing function with domain $\mathbb{N}_0$ and with $g \geq g^*$, and L an (s, Y_1, ℓ, r) -list-assignment of G . Let $\mathcal{T} = \{(A, B) : |V(A \cap B)| < \theta, |V(A) \cap Y_1| \leq 3\theta\}$. Assume that $\mathcal{T}$ is a tangle in G of order θ that does not control a $K_{t'}$ -minor.

Suppose that there exists no (η, g) -bounded L -coloring of G such that for every $x \in [s+3, s+2+r]$, the set of vertices colored x is a stable set, and suppose that for every separation $(A, B) \in \mathcal{T}$, every set Y_A with $|Y_A| \leq \eta^*(\theta)$ and $Y_1 \cap V(A) \subseteq Y_A \subseteq V(A)$ and every (s, Y_A, ℓ, r) -list-assignment L_A of $G[V(A)]$, there exists an (η, g) -bounded L_A -coloring of $G[V(A)]$ such that for every $x \in [s+3, s+2+r]$, the set of vertices colored x is a stable set.

Since $\mathcal{T}$ does not control a $K_{t'}$ -minor, by Theorem 21, there exist $Z \subseteq V(G)$ with $|Z| \leq \xi_0$, a $(\mathcal{T} - Z)$ -central segregation $\mathcal{S}$ of $G - Z$ of type (κ_0, ρ_0) , and a proper arrangement of $\mathcal{S}$ in a surface Σ in which $K_{t'}$ cannot be embedded. Let $\mathcal{S}_1 := \{(S, \Omega) \in \mathcal{S} : |\overline{\Omega}| \leq 3\}$, and let $\mathcal{S}_2 := \mathcal{S} - \mathcal{S}_1$. Since $\mathcal{S}$ is of type (κ_0, ρ_0) , $|\mathcal{S}_2| \leq \kappa_0$ and every member of $\mathcal{S}_2$ is a ρ_0 -vortex.

For each $(S, \Omega) \in \mathcal{S}_1$, let (A_S, B_S) be the separation of G such that $S \subseteq A_S$, $V(A_S) = V(S) \cup Z$, $V(A_S \cap B_S) = \overline{\Omega} \cup Z$, and subject to these conditions, $|E(A_S)|$ is minimal. Let $\mathcal{L}_1 := \{(A_S, B_S) : (S, \Omega) \in \mathcal{S}_1\}$. By Theorem 20, for each $(S, \Omega) \in \mathcal{S}_2$, there exists a vortical decomposition $(P_S, \mathcal{X}_S)$ of (S, Ω) of adhesion at most ρ_0 . For each $(S, \Omega) \in \mathcal{S}_2$ and each bag X of $(P_S, \mathcal{X}_S)$, let

$$\partial X := (X \cap \overline{\Omega}) \cup \{v \in X : v \text{ belongs to a bag of } (P_S, \mathcal{X}_S) \text{ other than } X\}.$$

Let $(A_{S,X}, B_{S,X})$ be the separation of G such that $V(A_{S,X}) := X \cup Z$ and $V(A_{S,X} \cap B_{S,X}) := \partial X \cup Z$, and subject to these conditions, $E(A_{S,X})$ is minimal. Let $\mathcal{L}_2 := \{(A_{S,X}, B_{S,X}) : (S, \Omega) \in \mathcal{S}_2, X \in \mathcal{X}_S\}$ and $\mathcal{L} := \mathcal{L}_1 \cup \mathcal{L}_2$. Note that every member of $\mathcal{L}$ has order at most $\xi_0 + 3 + 2\rho_0 < \theta^* \leq \theta$. Since $\mathcal{S}$ is $(\mathcal{T} - Z)$ -central, $\mathcal{L} \subseteq \mathcal{T}$. In addition, $\mathcal{L}$ is a location with $Z \subseteq V(A \cap B)$ for every $(A, B) \in \mathcal{L}$.

Let L' be a (Z, ℓ) -growth of L , and let $Y'_1 := \{v \in V(G) : |L'(v)| = 1\}$. By Lemma 22, $|Y'_1 \cap V(A)| \leq h(|V(A \cap B)| + |Y_1 \cap V(A)|) \leq h(4\theta)$ for every $(A, B) \in \mathcal{L}$. For each $(A, B) \in \mathcal{L}$, define:

$$U_A^{(0)} := (Y'_1 \cap V(A)) \cup V(A \cap B), \quad U_A^{(1)} := N_{G[V(A)]}^{\geq s}(U_A^{(0)}), \quad \text{and} \quad Z_A := U_A^{(0)} \cup U_A^{(1)}.$$

Note that for every $(A, B) \in \mathcal{L}$, $|Z_A| \leq f_1(h(4\theta) + \theta)$.

Let $G' = G[\bigcup_{(A,B) \in \mathcal{L}} Z_A]$. Since $Y'_1 \cap V(A) \subseteq Z_A$ for every $(A, B) \in \mathcal{L}$, we have $Y'_1 \subseteq V(G')$.

Claim 23.1. *There exists a Y'_1 -layering $\mathcal{V} = (V_1, V_2, \dots, V_{|\mathcal{V}|})$ of G' and a tree decomposition $(T, \mathcal{X})$ of G' with $\mathcal{V}$ -width at most $w_0(\theta)$ such that for every $(A, B) \in \mathcal{L}$, there exist $\epsilon_A \in \{1, -1\}$ and $i_A \in \mathbb{N}$ such that:*

- if $(A, B) \in \mathcal{L}_1$, then $V(A \cap B) - Y'_1 \subseteq V_{i_A} \cup V_{i_A + \epsilon_A}$ and $|V(A \cap B) \cap V_{i_A + \epsilon_A}| \leq 1$,
- if $(A, B) \in \mathcal{L}_2$, then $V(A \cap B) - Y'_1 \subseteq V_{i_A}$,
- $U_A^{(1)} \subseteq V_{i_A + \epsilon_A}$, and
- some bag of $(T, \mathcal{X})$ contains $U_A^{(1)}$.

Proof. Let G_0 be the graph obtained from $G[\bigcup_{(S,\Omega) \in \mathcal{S}} \overline{\Omega}]$ by for each $(S, \Omega) \in \mathcal{S}$, adding a new vertex v_S adjacent to all vertices in $\overline{\Omega}$ and new edges such that there exists a cycle C_S with $V(C_S) = \overline{\Omega}$ passing through its vertices in the order Ω . Since there exists a proper arrangement of $\mathcal{S}$ in Σ , G_0 can be embedded in Σ . Since $K_{t'}$ cannot be embedded in Σ , the Euler genus of Σ is at most σ . By [11, Theorem 12], there exists a layering $\mathcal{V}_0 = (V_{0,1}, V_{0,2}, \dots, V_{0,|\mathcal{V}_0|})$ of G_0 and a tree decomposition $(T_0, \mathcal{X}_0)$ of G_0 with $\mathcal{V}_0$ -width at most $2\sigma + 3$. For each $p \in V(T_0)$, let $X_{0,p}$ be the bag at p in $(T_0, \mathcal{X}_0)$.

For every $(S, \Omega) \in \mathcal{S}_1$, since $\{v_S\} \cup \overline{\Omega}$ forms a clique in G_0 , there exists $p_S \in V(T_0)$ such that X_{0,p_S} contains this clique. Let T'_0 be the tree obtained from T_0 by for each $(S, \Omega) \in \mathcal{S}_1$, adding a new node t_S and a new edge $t_S p_S$. For each $p \in V(T_0)$, let $X'_{0,p} :=$

$X_{0,p} - \{v_S : (S, \Omega) \in \mathcal{S}_1\}$. For each $p \in V(T'_0) - V(T_0)$, we know that $p = t_S$ for some $(S, \Omega) \in \mathcal{S}_1$; define $X'_{0,p} := \{v_S\} \cup \overline{\Omega}$. Since $N_{G_0}(v_S) = \overline{\Omega}$ for each $(S, \Omega) \in \mathcal{S}_1$, $(T'_0, \mathcal{X}'_0)$ is a tree decomposition of G_0 of $\mathcal{V}_0$ -width at most $2\sigma + 3$, where $\mathcal{X}'_0 = (X'_{0,p} : p \in V(T'_0))$.

For each $(S, \Omega) \in \mathcal{S}_1$, let i_S be the index such that $v_S \in V_{0,i_S}$; since $\{v_S\} \cup \overline{\Omega}$ forms a clique in G_0 , there exists $\epsilon_S \in \{-1, 1\}$ such that $\overline{\Omega} \subseteq V_{0,i_S} \cup V_{0,i_S+\epsilon_S}$. Let $\mathcal{V}'_0 := (V'_{0,1}, V'_{0,2}, \dots, V'_{0,|\mathcal{V}_0|})$ be the layering of G_0 obtained from $\mathcal{V}_0$ by moving v_S from V_{0,i_S} to $V_{0,i_S+\epsilon_S}$ for each $(S, \Omega) \in \mathcal{S}_1$ with $|\overline{\Omega} \cap V_{i_S+\epsilon_S}| > |\overline{\Omega} \cap V_{i_S}|$. Since for each $(S, \Omega) \in \mathcal{S}_1$, the only bag in $(T'_0, \mathcal{X}'_0)$ containing v_S has size $|\overline{\Omega}| + 1 \leq 4$, the $\mathcal{V}'_0$ -width of $(T'_0, \mathcal{X}'_0)$ is most $\max\{2\sigma + 3, 4\} \leq 2\sigma + 4$.

Let G_1 be the graph obtained from $G[\bigcap_{(A,B) \in \mathcal{L}} V(B)] - Z$ by adding a new vertex v_A adjacent to all vertices in $V(A \cap B) - Z$ for each $(A, B) \in \mathcal{L}_1$, and adding a new vertex v_S adjacent to all vertices in $\overline{\Omega}$ for each $(S, \Omega) \in \mathcal{S}_2$. Note that $V(G_0) \subseteq V(G_1)$. For each $i \in [|\mathcal{V}'_0|]$, let

$$V_{1,i} := V'_{0,i} \cup \bigcup_{(S,\Omega) \in \mathcal{S}_2, v_S \in V'_{0,i}} (V(G_1) \cap V(S) - \overline{\Omega}).$$

Let $\mathcal{V}_1 := (V_{1,1}, V_{1,2}, \dots, V_{1,|\mathcal{V}'_0|})$. Since $N_{G_1}(V(G_1) \cap V(S) - \overline{\Omega}) \subseteq \overline{\Omega} \subseteq N_{G_1}(v_S)$ for every $(S, \Omega) \in \mathcal{S}_2$, we have that $\mathcal{V}_1$ is a layering of G_1 . Note that for each $(S, \Omega) \in \mathcal{S}_2$ and $X \in \mathcal{X}_S$, X is the j -th bag in $\mathcal{X}_S$ for some j , so X contains the j -th vertex in Ω ; we denote the j -th vertex in Ω by v_X , so $v_X \in X$. For each node $p \in V(T'_0)$, define

$$X_{1,p} := X'_{0,p} \cup \bigcup_{(S,\Omega) \in \mathcal{S}_2} \bigcup_{X \in \mathcal{X}_S} \bigcup_{v_X \in X'_{0,p}} (X \cap V(G_1)).$$

Since $(T'_0, \mathcal{X}'_0)$ is a tree decomposition of G_0 and $(P_S, \mathcal{X}_S)$ is a vortical decomposition of S for every $(S, \Omega) \in \mathcal{S}_2$, $(T'_0, \mathcal{X}_1)$ is a tree decomposition of G_1 by the existence of C_S , where for every $p \in V(T'_0)$, the bag of $(T'_0, \mathcal{X}_1)$ at p is $X_{1,p}$. Since $(P_S, \mathcal{X}_S)$ has adhesion at most ρ_0 for each $(S, \Omega) \in \mathcal{S}_2$, the $\mathcal{V}_1$ -width of $(T'_0, \mathcal{X}_1)$ is at most $(2\sigma + 4)(2\rho_0 + 1)$.

For each $(S, \Omega) \in \mathcal{S}_2$, since v_S is adjacent to all vertices in $\overline{\Omega}$, there exists $i_S \in [|\mathcal{V}_1|]$ such that $v_S \in V_{1,i_S}$, and $\overline{\Omega}$ is contained in $V_{1,i_S-1} \cup V_{1,i_S} \cup V_{1,i_S+1}$. Let $\mathcal{V}'_1$ be the layering of G_1 obtained from $\mathcal{V}_1$ by for each $(S, \Omega) \in \mathcal{S}_2$, merging $V_{1,i_S-1} \cup V_{1,i_S} \cup V_{1,i_S+1}$ into a layer. Since $|\mathcal{S}_2| \leq \kappa_0$, each layer in $\mathcal{V}'_1$ is a union of at most $3\kappa_0$ layers in $\mathcal{V}_1$. Hence the $\mathcal{V}'_1$ -width of $(T'_0, \mathcal{X}_1)$ is at most $3\kappa_0(2\sigma + 4)(2\rho_0 + 1)$. Denote $\mathcal{V}'_1 = (V'_{1,1}, V'_{1,2}, \dots, V'_{1,|\mathcal{V}'_1|})$.

For each $(A, B) \in \mathcal{L}_1$, let i_A be the index i such that $v_A \in V'_{1,i}$; for each $(A, B) \in \mathcal{L}_2$, let i_A be the index i such that $v_S \in V'_{1,i}$, where (S, Ω) is the member in $\mathcal{S}_2$ such that $V(A) - Z \subseteq V(S)$. Note that by the definition of $\mathcal{V}'_1$, for each $(A, B) \in \mathcal{L}_2$, $V(A \cap B) - Z \subseteq V'_{1,i_A}$. By adding empty layers, we may assume that $|\mathcal{V}'_1| \geq 2 + \max_{(A,B) \in \mathcal{L}} i_A$ and $\min_{(A,B) \in \mathcal{L}} i_A \geq 3$.

For each $(A, B) \in \mathcal{L}_1$, since $|V(A \cap B) - Z| \leq 3$, we have that $(V(A \cap B) - Z) \cup \{v_A\}$ forms a clique in G_1 ; since $v_A \in V'_{1,i_A}$, there exists $\epsilon_A \in \{1, -1\}$ such that $V(A \cap B) - Z \subseteq V'_{1,i_A} \cup V'_{1,i_A+\epsilon_A}$. Note that by the definition of $\mathcal{V}'_0$ and $\mathcal{V}'_1$, for each $(A, B) \in \mathcal{L}_1$,

$|V(A \cap B) \cap V'_{1,i_A}| \geq |V(A \cap B) \cap V'_{1,i_A+\epsilon_A}|$, so $|V(A \cap B) \cap V'_{1,i_A+\epsilon_A}| \leq \lfloor \frac{|V(A \cap B) - Z|}{2} \rfloor \leq \lfloor \frac{3}{2} \rfloor = 1$. For each $(A, B) \in \mathcal{L}_2$, define $\epsilon_A := 1$.

For each $j \in [|\mathcal{V}'_1|]$, define

$$V_{2,j} := (V'_{1,j} \cup \bigcup_{(A,B) \in \mathcal{L}, i_A+\epsilon_A=j} U_A^{(1)}) \cap V(G').$$

Let $\mathcal{V}_2 := (V_{2,1}, V_{2,2}, \dots, V_{2,|\mathcal{V}'_1|})$. For each $i \in [|\mathcal{V}_2|]$, let $V_i := V_{2,i} - Y'_1$. Define $\mathcal{V} := (V_1, V_2, \dots, V_{|\mathcal{V}_2|})$.

Note that for each $(A, B) \in \mathcal{L}$, we have $V(A \cap B) - Y'_1 \subseteq (V'_{1,i_A} \cup V'_{1,i_A+\epsilon_A}) - Y'_1 \subseteq (V_{2,i_A} \cup V_{2,i_A+\epsilon_A}) - Y'_1 \subseteq V_{i_A} \cup V_{i_A+\epsilon_A}$ and $|V(A \cap B) \cap V_{i_A+\epsilon_A}| \leq |V(A \cap B) \cap V'_{1,i_A+\epsilon_A}| \leq 1$. Moreover, if $(A, B) \in \mathcal{L}_2$, then $V(A \cap B) - Y'_1 \subseteq V'_{1,i_A} - Y'_1 \subseteq V_{2,i_A} - Y'_1 = V_{i_A}$. For every $(A, B) \in \mathcal{L}$, since $U_A^{(1)} \subseteq V(A) - V(B)$, we have $N_{G'}(U_A^{(1)}) - Y'_1 \subseteq V(A \cap B) - Y'_1 \subseteq (V_{i_A} \cup V_{i_A+\epsilon_A}) - Y'_1$. So $\mathcal{V}$ is a Y'_1 -layering of G' since $U_A^{(1)} \subseteq V_{i_A+\epsilon_A}$ for every $(A, B) \in \mathcal{L}$.

For each $p \in V(T'_0)$, define

$$X_{2,p} := (X_{1,p} \cup \bigcup_{(A,B) \in \mathcal{L}_1, v_A \in X_{1,p}} (Z_A - V(G_1)) \cup \bigcup_{(A,B) \in \mathcal{L}_2, V(A \cap B) - Z \subseteq X_{1,p}, V(A \cap B) - Z \neq \emptyset} (Z_A - V(G_1))) \cap V(G').$$

Let p_0 be a node of T'_0 . Let T be the tree obtained from T'_0 by adding, for each $(A, B) \in \mathcal{L}_2$ with $V(A \cap B) - Z = \emptyset$, a new node t_A and a new edge $t_A p_0$. For each $p \in V(T'_0)$, let $X_p := X_{2,p} \cup Y'_1$; for each $p \in V(T) - V(T'_0)$, we know $p = t_A$ for some $(A, B) \in \mathcal{L}_2$ with $V(A \cap B) - Z = \emptyset$; let $X_p := (Z_A - V(G_1)) \cup Y'_1$. Let $\mathcal{X} := (X_p : p \in V(T))$.

Note that for every $(A, B) \in \mathcal{L}_1$, we have $N_{G'}(Z_A - V(G_1)) \subseteq N_{G_1}(v_A) \cup Y'_1$; for every $(A, B) \in \mathcal{L}_2$, we have $N_{G'}(Z_A - V(G_1)) \subseteq V(A \cap B) \cup Y'_1$. So $(T, \mathcal{X})$ is a tree decomposition of G' . For every $(A, B) \in \mathcal{L}$, since $U_A^{(1)} \subseteq Z_A - V(G_1)$, some bag of $(T, \mathcal{X})$ contains $U_A^{(1)}$.

Since $\mathcal{V}$ is a Y'_1 -layering, and $|Z_A| \leq f_1(h(4\theta) + \theta)$ for every $(A, B) \in \mathcal{L}$, the $\mathcal{V}$ -width of $(T, \mathcal{X})$ is at most $(3\kappa_0(2\sigma + 4)(2\rho_0 + 1) + 1) \cdot (f_1(h(4\theta) + \theta) + 1) \leq w_0(\theta)$. This proves the claim. $\square$

Let $\mathcal{V} := (V_1, V_2, \dots, V_{|\mathcal{V}|})$ be the Y'_1 -layering of G' mentioned in Claim 23.1. For every $(A, B) \in \mathcal{L}$, let i_A and ϵ_A be the integers mentioned in Claim 23.1.

Let G^* be the graph obtained from G' by:

- adding $s + 2$ new vertices $z_1^*, z_2^*, \dots, z_{s+2}^*$,
- for each $(A, B) \in \mathcal{L}$ and each 2-element subset S of $[2, s + 2]$,
 - adding a new vertex $u_{A,S}^*$ and all edges between $u_{A,S}^*$ and $U_A^{(1)}$, and
 - adding new edges $u_{A,S}^* z_i^*$ for all $i \in [s + 2]$ with $i \not\equiv i_A + j\epsilon_A \pmod{s + 2}$, where $j \in S \cup \{1\}$.

Let $Z^* = \{z_i^* : i \in [s+2]\}$. For each $(A, B) \in \mathcal{L}$, let $U_A^* = \{u_{A,S}^* : S \subseteq [2, s+2], |S| = 2\}$.

Claim 23.2. G^* does not contain a $K_{s,t+f_1(h(4\theta)+\theta)+(s+2)^2}$ -subgraph.

Proof. Suppose to the contrary that some subgraph Q of G^* is isomorphic to $K_{s,t+f_1(h(4\theta)+\theta)+(s+2)^2}$. Let $\{S, T\}$ be a bipartition of $V(Q)$ with $|S| = s$.

If there exist $(A, B) \in \mathcal{L}$ and $v \in S \cap (U_A^* \cup (V(A) - V(B))) \neq \emptyset$, then since $N_{G^*}(v) \subseteq Z_A \cup U_A^* \cup Z^*$, v is adjacent in G^* to at most $f_1(h(4\theta) + \theta) + \binom{s+1}{2} + (s+2) < f_1(h(4\theta) + \theta) + (s+2)^2$ vertices, so $v \notin S$, a contradiction.

So $S \cap (U_A^* \cup (V(A) - V(B))) = \emptyset$ for every $(A, B) \in \mathcal{L}$. Suppose that there exist $(A, B) \in \mathcal{L}$ and $u \in U_A^* \cap V(Q)$. Then $u \in T$. Since u is adjacent to at most $s+2-3 = s-1$ vertices in Z^* , $U_A^{(1)} \cap S \neq \emptyset$. But $U_A^{(1)} \subseteq V(A) - V(B)$, a contradiction.

So there does not exist $(A, B) \in \mathcal{L}$ with $U_A^* \cap V(Q) \neq \emptyset$. Hence if some vertex in Z^* is in $V(Q)$, then it has no neighbor in Q , a contradiction. So $V(Q) \cap Z^* = \emptyset$. Therefore, Q is a subgraph of G' , a contradiction. $\square$

For each $i \in [|\mathcal{V}|]$, let $V_i^* = V_i \cup \bigcup_{(A,B) \in \mathcal{L}, U_A^{(1)} \subseteq V_i} U_A^*$. Let $\mathcal{V}^* = (V_1^*, V_2^*, \dots, V_{|\mathcal{V}|}^*)$. Note that $\mathcal{V}^*$ is a $(Y_1' \cup Z^*)$ -layering of G^* .

Claim 23.3. There exists a tree decomposition of G^* with $\mathcal{V}^*$ -width at most $w_0(\theta) \cdot (s+2)^2$.

Proof. Let $(T, \mathcal{X})$ be the tree decomposition of G' with $\mathcal{V}$ -width at most $w_0(\theta)$ mentioned in Claim 23.1. Add Z^* into all bags of $(T, \mathcal{X})$. For each $(A, B) \in \mathcal{L}$, Claim 23.1 implies that there exists a node p_A of T such that the bag at p_A contains $U_A^{(1)}$. If $U_A^{(1)} \neq \emptyset$, then add U_A^* into the bag at p_A ; if $U_A^{(1)} = \emptyset$, then add a new node into T adjacent to p_A with bag $Z^* \cup U_A^*$. We obtain a tree decomposition of G^* with $\mathcal{V}^*$ -width at most $w_0(\theta) \cdot (s+2)^2$. $\square$

Let $L_{\mathcal{L}}$ be the following list-assignment of G^* :

- For every $v \in V(G') - Y_1'$, let $L_{\mathcal{L}}(v) := L'(v) \cap [s+2] - \{i\}$, where $i \in [s+2]$ is the number such that $v \in V_j$ and $j \equiv i \pmod{s+2}$.
- For every $v \in Y_1'$, let $L_{\mathcal{L}}(v) := L'(v) \cap [s+2]$.
- For every $i \in [s+2]$, let $L_{\mathcal{L}}(z_i^*) = \{i\}$.
- For every $(A, B) \in \mathcal{L}$ and $S \subseteq [2, s+2]$ with $|S| = 2$, let $L_{\mathcal{L}}(u_{A,S}^*) = \{i \in [s+2] : i \equiv i_A + j_{\in A} \pmod{s+2} \text{ with } j \in S\}$.

Let G'' be the subgraph of G^* induced by $\{v \in V(G^*) : L_{\mathcal{L}}(v) \neq \emptyset\}$. Note that for every $y \in Y_1'$, $y \in V(G'')$ if and only if $L'(y) \in [s+2]$.

Claim 23.4. For every $v \in V(G') - Y_1'$,

$$|L_{\mathcal{L}}(v)| \geq |L'(v)| - r + |N_G(v) \cap Y_1'| - |N_G(v) \cap Y_1' \cap V(G'')| - 1.$$

Proof. Since L' satisfies (R5), for every $v \in V(G') - Y'_1$,

$$\begin{aligned} |L_{\mathcal{L}}(v)| &\geq |L'(v)| - |L'(v) \cap [s+3, s+2+r]| - 1 \\ &= |L'(v)| - (r - |\{y \in N_G(v) \cap Y'_1 : L'(y) \subseteq [s+3, s+r+2]\}|) - 1 \\ &= |L'(v)| - (r - |N_G(v) \cap Y'_1 - V(G'')|) - 1 \\ &= |L'(v)| - r + |N_G(v) \cap Y'_1| - |N_G(v) \cap Y'_1 \cap V(G'')| - 1. \quad \square \end{aligned}$$

For every $v \in V(G') - Y'_1$, since L' is an $(s, r+2, Y'_1)$ -list-assignment, we know $|L'(v)| \geq r+3$, so $|L_{\mathcal{L}}(v)| \geq 2$ by Claim 23.4, and hence $v \in V(G'')$. So $V(G^*) - V(G'') \subseteq Y'_1$.

Let $Y''_1 = (V(G'') \cap Y'_1) \cup Z^*$. Note that $Y''_1 = \{y \in V(G'') : |L_{\mathcal{L}}(y)| = 1\}$.

Claim 23.5. *For every $y \in V(G'') \cap N_G^{<s}(Y''_1)$, there exists a subset $L_{\mathcal{L}, G''}(y)$ of $L_{\mathcal{L}}(y)$ with $|L_{\mathcal{L}, G''}(y)| = s+1 - |N_{G''}(y) \cap Y''_1|$ such that $L_{\mathcal{L}, G''}(y) \cap L_{\mathcal{L}}(u) = \emptyset$ for every $u \in N_{G''}(y) \cap Y''_1$.*

Proof. Let $y \in V(G'') \cap N_G^{<s}(Y''_1)$.

First suppose $y \in N_{G''}(Z^*)$. Then $y \in U_A^*$ for some $(A, B) \in \mathcal{L}$, so $|L_{\mathcal{L}}(y)| = 2$ and $y = u_{A,S}^*$ for some $S \subseteq [2, s+2]$ with $|S| = 2$. Hence $N_{G''}(y) \cap Y''_1 = N_{G^*}(y) \cap Z^* = \{z_i^* : i \in [s+2], i \neq i_A + j \in A, j \in S \cup \{1\}\}$. So $|N_{G''}(y) \cap Y''_1| = s+2-3 = s-1$ and $L_{\mathcal{L}}(y) \cap L_{\mathcal{L}}(u) = \emptyset$ for every $u \in N_{G''}(y) \cap Y''_1$. Hence $L_{\mathcal{L}}(y)$ is a desired subset $L_{\mathcal{L}, G''}(y)$ of $L_{\mathcal{L}}(y)$.

Now assume $y \notin N_{G''}(Z^*)$. Hence $y \in V(G')$ and $N_{G''}(y) \cap Y''_1 = N_{G''}(y) \cap Y'_1 \cap V(G'') = N_G(y) \cap Y'_1 \cap V(G'')$. If $y \in N_G^{<s}(Y'_1) \cup (V(G) - N_G[Y'_1])$, then since L' is an $(s, r+2, Y'_1)$ -list-assignment of G , by Claim 23.4,

$$\begin{aligned} |L_{\mathcal{L}}(y)| &\geq |L'(y)| - r + |N_G(y) \cap Y'_1| - |N_G(y) \cap Y'_1 \cap V(G'')| - 1 \\ &= (s+r+2 - |N_G(y) \cap Y'_1|) - r + |N_G(y) \cap Y'_1| - |N_G(y) \cap Y'_1 \cap V(G'')| - 1 \\ &= s+1 - |N_G(y) \cap Y'_1 \cap V(G'')| \\ &= s+1 - |N_{G''}(y) \cap Y''_1|, \end{aligned}$$

and $L_{\mathcal{L}}(y) \cap L_{\mathcal{L}}(u) \subseteq L'(y) \cap L'(u) = \emptyset$ for every $u \in N_G(y) \cap Y'_1 \cap V(G'') = N_{G''}(y) \cap Y''_1$, so a desired subset $L_{\mathcal{L}, G''}(y)$ of $L_{\mathcal{L}}(y)$ exists.

So we may assume $y \in V(G') - (N_G^{<s}(Y'_1) \cup (V(G) - N_G[Y'_1]))$. Hence $y \in N_G^{\geq s}(Y'_1)$. In particular, $y \notin Y'_1$, so $y \notin N_G^{\geq s}(Y'_1 \cup Z)$ since L' is a (Z, ℓ) -growth of L . For each $i \in [0, s+2]$, let $L^{(i)}$ and $Y_1^{(i)}$ be the list-assignment $L^{(i)}$ and the subset $Y_1^{(i)}$ of $V(G)$ mentioned in the definition of a (Z, ℓ) -growth. Since $y \in N_G^{\geq s}(Y'_1) - N_G^{\geq s}(Y'_1 \cup Z) = N_G^{\geq s}(Y_1^{(s+2)}) - N_G^{\geq s}(Y_1^{(0)})$, there exists $j \in [s+2]$ such that $y \in N_G^{\geq s}(Y_1^{(j)}) - N_G^{\geq s}(Y_1^{(j-1)})$. So $|L'(y)| = |L^{(j)}(y)| = |L^{(j-1)}(y)| \geq s+r+2 - |N_G(y) \cap Y_1^{(j-1)}|$, and for every $u \in N_G(y) \cap Y_1^{(j-1)}$, $L_{\mathcal{L}}(y) \cap L_{\mathcal{L}}(u) \subseteq L'(y) \cap L'(u) \subseteq L^{(j-1)}(y) \cap L^{(j-1)}(u) = \emptyset$. Hence by Claim 23.4, since $Y'_1 \supseteq Y_1^{(j-1)}$,

$$\begin{aligned}
|L_{\mathcal{L}}(y)| &\geq |L'(y)| - r + |N_G(y) \cap Y_1'| - |N_G(y) \cap Y_1' \cap V(G'')| - 1 \\
&\geq (s + r + 2 - |N_G(y) \cap Y_1^{(j-1)}|) - r \\
&\quad + |N_G(y) \cap Y_1'| - |N_G(y) \cap Y_1' \cap V(G'')| - 1 \\
&= s + 1 - |N_G(y) \cap Y_1' \cap V(G'')| + (|N_G(y) \cap Y_1'| - |N_G(y) \cap Y_1^{(j-1)}|) \\
&= s + 1 - |N_{G''}(y) \cap Y_1''| + (|N_G(y) \cap Y_1' - Y_1^{(j-1)}|).
\end{aligned}$$

So there exists a subset $L_{\mathcal{L},G''}(y)$ of $L_{\mathcal{L}}(y)$ with size $s + 1 - |N_{G''}(y) \cap Y_1''|$ such that $L_{\mathcal{L},G''}(y) \cap L_{\mathcal{L}}(u) = \emptyset$ for every $u \in N_G(y) \cap Y_1' - Y_1^{(j-1)}$. Hence $L_{\mathcal{L},G''}(y) \cap L_{\mathcal{L}}(u) = \emptyset$ for every $u \in N_G(y) \cap Y_1' \cap V(G'') = N_{G''}(y) \cap Y_1''$. $\square$

Claim 23.6. *There exists an $(s, 1, Y_1'')$ -list-assignment $L_{\mathcal{L},G''}$ of G'' such that $L_{\mathcal{L},G''}(v) \subseteq L_{\mathcal{L}}(v)$ for every $v \in V(G'')$.*

Proof. For every $v \in Y_1''$, let $L_{\mathcal{L},G''}(v) = L_{\mathcal{L}}(v)$. For every $v \in V(G'') \cap N_{G''}^{\leq s}(Y_1'')$, let $L_{\mathcal{L},G''}(v)$ be the set $L_{\mathcal{L},G''}(v)$ mentioned in Claim 23.5. So $L_{\mathcal{L},G''}$ satisfies (L3) by Claim 23.5.

For every $v \in V(G'') - N_{G''}[Y_1'']$ with $v \in V(G'') - N_G[Y_1']$, we know either $v \in V(G') - Y_1'$, or $s = 1$ and $v \in \bigcup_{(A,B) \in \mathcal{L}} U_A^*$; for the former, by Claim 23.4,

$$\begin{aligned}
|L_{\mathcal{L}}(v)| &\geq |L'(v)| - r + |N_G(v) \cap Y_1'| - |N_G(v) \cap Y_1' \cap V(G'')| - 1 \\
&= |L'(v)| - r - 1 \\
&\geq (s + r + 2) - r - 1 \\
&= s + 1;
\end{aligned}$$

for the latter, $|L_{\mathcal{L}}(v)| = 2 = s + 1$. So there exists $L_{\mathcal{L},G''}(v) \subseteq L_{\mathcal{L}}(v)$ with $|L_{\mathcal{L},G''}(v)| = s + 1$ in each case.

Let $v \in V(G'') - N_{G''}[Y_1'']$ with $v \notin V(G'') - N_G[Y_1']$. So $v \in V(G'') \cap N_G[Y_1'] - N_{G''}[Y_1'']$. Hence $v \in V(G') - Y_1'$, and for every $y \in N_G(v) \cap Y_1'$, $L'(y) \in [s + 3, s + r + 2]$ and hence $y \in Y_1$ (since L' is a (Z, ℓ) -growth of L). So $N_G(v) \cap Y_1' = N_G(v) \cap Y_1$. This implies $L'(v) = L(v)$ since L' is a (Z, ℓ) -growth of L .

Since $v \in N_G[Y_1'] - Y_1'$, we know $v \in N_G^{\leq s}(Y_1') \cup N_G^{\geq s}(Y_1')$. Since $N_G(v) \cap Y_1' = N_G(v) \cap Y_1$, $v \in N_G^{\leq s}(Y_1) \cup N_G^{\geq s}(Y_1)$. Since $v \notin Y_1'$, $v \notin N_G^{\geq s}(Y_1)$. So $v \in N_G^{\leq s}(Y_1)$. Hence $|L(v)| = s + r + 2 - |N_G(v) \cap Y_1|$. By Claim 23.4,

$$\begin{aligned}
|L_{\mathcal{L}}(v)| &\geq |L'(v)| - r + |N_G(v) \cap Y_1'| - |N_G(v) \cap Y_1' \cap V(G'')| - 1 \\
&= |L(v)| - r + |N_G(v) \cap Y_1| - |N_G(v) \cap Y_1' \cap V(G'')| - 1 \\
&= (s + r + 2 - |N_G(v) \cap Y_1|) - r + |N_G(v) \cap Y_1| - |N_G(v) \cap Y_1' \cap V(G'')| - 1 \\
&= s + 1 - |N_G(v) \cap Y_1' \cap V(G'')|.
\end{aligned}$$

Since $v \notin N_{G''}[Y_1'']$, $|N_G(v) \cap Y_1' \cap V(G'')| = 0$. So $|\mathcal{L}_{\mathcal{L}}(v)| \geq s+1$. Therefore, there exists $L_{\mathcal{L},G''}(v) \subseteq L_{\mathcal{L}}(v)$ with $|L_{\mathcal{L},G''}(v)| = s+1$. So $L_{\mathcal{L},G''}$ satisfies (L4).

Moreover, for every $x \in V(G'') - Y_1''$ in which $L_{\mathcal{L},G''}(x)$ has not been defined, since $|L_{\mathcal{L}}(x)| \geq 2$ for every $x \in V(G'') - Y_1''$, there exists $L_{\mathcal{L},G''}(x) \subseteq L_{\mathcal{L}}(x)$ with $|L_{\mathcal{L},G''}(x)| = 2$. This implies that $L_{\mathcal{L},G''}$ is an $(s, 1, Y_1'')$ -list-assignment of G'' and proves the claim. $\square$

Let $L_{\mathcal{L},G''}$ be an $(s, 1, Y_1'')$ -list-assignment of G'' stated in Claim 23.6.

Recall that $V(G^*) - V(G'') \subseteq Y_1'$ and $\mathcal{V}^*$ is a $(Y_1' \cup Z^*)$ -layering of G^* . So $\mathcal{V}^*$ is a Y_1'' -layering of G'' . Note that $L_{\mathcal{L}}|_{V(G'')}$ is $(s, \mathcal{V}^*)$ -compatible. So $L_{\mathcal{L},G''}$ is $(s, \mathcal{V}^*)$ -compatible. Since $L_{\mathcal{L},G''}$ is an $(s, 1, Y_1'')$ -list-assignment of G'' , $(Y_1'', L_{\mathcal{L},G''})$ is a $\mathcal{V}^*$ -standard pair.

Claim 23.7. *There exists an $(L' \cup L_{\mathcal{L}})$ -coloring $c_{\mathcal{L}}$ of G^* with clustering $\eta_1(\eta + \theta)$ such that:*

- $c_{\mathcal{L}}|_{V(G'')}$ is an $L_{\mathcal{L},G''}$ -coloring of G'' ,
- for every $x \in [s+3, s+2+r]$, $\{v \in V(G^*) : c_{\mathcal{L}}(v) = x\} = \{v \in Y_1' : L'(v) = \{x\}\}$ is a stable set, and
- for every $(A, B) \in \mathcal{L}$, there exists at most one element k_A in $[s+r+2]$ such that there exist two monochromatic components with respect to $c_{\mathcal{L}}$ each having color k_A and intersecting $U_A^{(1)}$.

Proof. Since G'' is an induced subgraph of G^* , by Claims 23.2 and 23.3, G'' does not contain a $K_{s,t+f_1(h(4\theta)+\theta)+(s+2)^2}$ subgraph, and there exists a tree-decomposition of G'' with $\mathcal{V}^*$ -width at most $(s+2)^2 \cdot w_0(\theta)$. By Lemma 22, $|Y_1''| \leq |Y_1'| + |Z^*| \leq h(|Y_1'| + |Z|) + s+2 \leq h(\eta + \xi_0) + s+2$. Since $L_{\mathcal{L},G''}$ is $(s, \mathcal{V}^*)$ -compatible and $(Y_1'', L_{\mathcal{L},G''})$ is a $\mathcal{V}^*$ -standard pair, by Theorem 15, there exists an $L_{\mathcal{L},G''}$ -coloring $c_{\mathcal{L}}$ of G'' with clustering $\eta_1(\eta + \theta)$. By further coloring each vertex y in $Y_1' - V(G'')$ with the unique element in $L'(y)$, we extend the coloring $c_{\mathcal{L}}$ to be an $(L' \cup L_{\mathcal{L}})$ -coloring of G^* with clustering $\eta_1(\eta + \theta)$, and $\{v \in V(G^*) : c_{\mathcal{L}}(v) = x\} = \{v \in Y_1' : L'(v) = \{x\}\}$ is a stable set for every $x \in [s+3, s+2+r]$.

Let $(A, B) \in \mathcal{L}$. Let M_1 and M_2 be two distinct monochromatic components with respect to $c_{\mathcal{L}}$ having the same color and intersecting $U_A^{(1)}$. Let k be the color of M_1 and M_2 . Since M_1 intersects $U_A^{(1)} \subseteq V_{i_A+\epsilon_A}$, we know $k \in [s+2]$, and by the definition of $L_{\mathcal{L}}$, $k \not\equiv i_A + \epsilon_A \pmod{s+2}$. That is, $k \equiv i_A + j\epsilon_A \pmod{s+2}$ for some $j \in [2, s+2]$. Since every vertex in U_A^* is adjacent to every vertex in $U_A^{(1)}$, no vertex in U_A^* is colored with k by $c_{\mathcal{L}}$, for otherwise, M_1 and M_2 are not distinct. Hence, if there are two possibilities for k , then there exists $S \subseteq [2, s+2]$ with $|S| = 2$ such that $u_{A,S}^*$ cannot be colored by $c_{\mathcal{L}}$ by the definition of $L_{\mathcal{L}}(u_{A,S}^*)$, a contradiction. $\square$

Let $c_{\mathcal{L}}$ be the $(L' \cup L_{\mathcal{L}})$ -coloring of G^* mentioned in Claim 23.7.

Claim 23.8. *For each $(A, B) \in \mathcal{L}$, there exist $Y_A \subseteq V(A)$ with $Z_A \subseteq Y_A$ and with $|Y_A| \leq \eta^*(\theta)$ and an (s, Y_A, ℓ, r) -list-assignment L_A of $G[V(A)]$ such that:*

- $L_A(v) = \{c_{\mathcal{L}}(v)\}$ for every $v \in Z_A$,
- $L_A(v) \subseteq L'(v)$ for every $v \in V(A)$, and
- for every L_A -coloring c_A of $G[V(A)]$, every monochromatic component M of $G[V(A)]$ with respect to c_A intersecting $(Y'_1 \cap V(A)) \cup V(A \cap B) = U_A^{(0)}$ either:
 - is contained in $G[Z_A]$, or
 - contains a vertex in $Z_A - V(B)$, is contained in $G[Y_A]$, and some monochromatic component of G^* with respect to $c_{\mathcal{L}}$ contains $M[V(M) \cap Z_A]$.

Proof. For every separation (A, B) , define the following:

- Let $Y'_A := Z_A$.
- For every $v \in Y'_A$, let $L'_A(v) := \{c_{\mathcal{L}}(v)\}$.
- For every $v \in V(A) - Y'_A$ with $1 \leq |N_{G[V(A)]}(v) \cap Y'_A| \leq s - 1$, let $L'_A(v)$ be a subset of $L'(v)$ with size $s + r + 2 - |N_{G[V(A)]}(v) \cap Y'_A|$ such that $L'_A(v) \cap L'_A(y) = \emptyset$ for every $y \in N_{G[V(A)]}(v) \cap Y'_A$, and $|\{y \in N_G(v) \cap Y'_A : L'_A(y) \subseteq [s + 3, s + 2 + r]\}| = r - |L'_A(v) \cap [s + 3, s + 2 + r]|$. (Such $L'_A(v)$ exists since $v \in V(A) - Y'_A$ implies $v \in V(A) - V(B)$, and L' satisfies (R5).)
- For every $v \in V(A) - Y'_A$ with $|N_{G[V(A)]}(v) \cap Y'_A| \geq s$ and $1 \leq |N_{G[V(A)]}(v) \cap U_A^{(0)}| \leq s - 1$, let $L'_A(v)$ be a subset of $L'(v)$ with size $s + r + 2 - |N_{G[V(A)]}(v) \cap U_A^{(0)}|$ such that $L'_A(v) \cap L'_A(y) = \emptyset$ for every $y \in N_{G[V(A)]}(v) \cap U_A^{(0)}$, and $|\{y \in N_G(v) \cap Y'_A : L'_A(y) \subseteq [s + 3, s + 2 + r]\}| = r - |L'_A(v) \cap [s + 3, s + 2 + r]|$. (Such $L'_A(v)$ exists since $v \in V(A) - Y'_A$ implies $v \in V(A) - V(B)$, and $Y'_1 \cap V(A) \subseteq U_A^{(0)}$ and L' satisfies (R5).)
- For every other vertex v in $V(A)$, let $L'_A(v) := L'(v)$.

Since $V(A \cap B) \subseteq Y'_A$, and L' is an $(s, r + 2, Y'_1)$ -list-assignment of G , L'_A is an $(s, r + 2, Y'_A)$ -list-assignment of $G[V(A)]$. Since $\{v \in V(G^*) : c_{\mathcal{L}}(v) = x\} = \{v \in Y'_1 : L'(v) = \{x\}\}$ is a stable set for every $x \in [s + 3, s + 2 + r]$, and L' satisfies (R3) and (R4) (with Y_1 replaced by Y'_1), we know L'_A satisfies (R4) (with Y_1 replaced by Y'_A), and $\{v \in N_{G[V(A)]}(y) - Y'_A : x \in L'_A(v)\} = \emptyset$ for every $y \in Y'_A$ and $x \in [s + 3, s + 2 + r]$ with $x \in L'_A(y)$. Since $Y'_1 \cap V(A) \subseteq U_A^{(0)}$ and $N_{G[V(A)]}^{\geq s}(U_A^{(0)}) \subseteq U_A^{(1)} \subseteq Y'_A$ and L' satisfies (R5) (with Y_1 replaced by Y'_1), L'_A satisfies (R5) (with Y_1 replaced by Y'_A) by the definition of L'_A . In addition, $|Y'_A| = |Z_A| \leq f_1(h(4\theta) + \theta)$.

For each $(A, B) \in \mathcal{L}$, if the element k_A in $[s + r + 2]$ mentioned in Claim 23.7 exists, then let k_A be this element; otherwise, let $k_A = 0$. Note that $k_A \notin [s + 3, s + r + 2]$ since if k_A exists in Claim 23.7, then the corresponding monochromatic component intersects $U_A^{(1)}$.

For each $(A, B) \in \mathcal{L}$, we further define the following:

- Let $W_A := \{v \in V(A) - Y'_A : |N_{G[V(A)]}(v) \cap Y'_A| \geq s\}$.
- Let $L'_{A,W}$ be a $(W_A, \{k_A\} \cup [s + 3, s + 2 + r])$ -progress of L'_A (in $G[A]$).
- Let $W'_A = \{v \in V(A) - (Y'_A \cup W_A) : |N_{G[V(A)]}(v) \cap (Y'_A \cup W_A)| \geq s\}$.

- Let L''_A be a $(W'_A, \{\ell\} \cup [s+3, s+2+r])$ -progress of $L'_{A,W}$ (in $G[A]$).
- Let $Y''_A := Y'_A \cup W_A \cup W'_A$.

Since L'_A is an $(s, r+2, Y'_A)$ -list-assignment of $G[V(A)]$, we have L''_A satisfies (R1)–(R3) (with Y_1 replaced by Y''_A) by Lemma 12. By the definition of L''_A , for every $x \in [s+3, s+2+r]$, we have $\{y \in Y''_A : x \in L''_A(y)\} = \{y \in Y'_A : x \in L'_A(y)\}$, so L''_A satisfies (R4) and (R5) (with Y_1 replaced by Y''_A) by Lemma 12. Hence L''_A is an (s, Y''_A, ℓ, r) -list-assignment of $G[V(A)]$. Note that by Lemma 11,

$$|Y''_A| \leq |Y'_A| + |W_A| + |W'_A| \leq |Y'_A| + f(|Y'_A|) + f(|Y'_A| + f(|Y'_A|)) \leq f_3(h(4\theta) + \theta).$$

For each $(A, B) \in \mathcal{L}$, define L_A to be a (Y''_A, ℓ) -growth of L''_A (in $G[V(A)]$), and let $Y_A = \{v \in V(A) : |L_A(v)| = 1\}$. By Lemma 22, L_A is an (s, Y_A, ℓ, r) -list-assignment of $G[V(A)]$. Furthermore, it is clear that $L_A(v) \subseteq L''_A(v) \subseteq L'_A(v) \subseteq L'(v)$ for every $v \in V(A)$, so $L_A(v) = \{c_{\mathcal{L}}(v)\}$ for every $v \in Y'_A$, and every L_A -coloring is an L'_A -coloring and an L''_A -coloring. By Lemma 22,

$$|Y_A| \leq h(|Y''_A|) \leq h(f_3(h(4\theta) + \theta)) = \eta^*(\theta).$$

In addition, for every $(A, B) \in \mathcal{L}$, we have $Y_1 \cap V(A) \subseteq Y'_A \subseteq Y''_A \subseteq Y_A \subseteq V(A)$.

Now we prove the last statement of this claim. Assume that (A, B) is a fixed element of $\mathcal{L}$ and M is a fixed monochromatic component with respect to a fixed L_A -coloring c_A of $G[V(A)]$ intersecting $(Y'_1 \cap V(A)) \cup V(A \cap B) = U_A^{(0)}$. We may assume that M is not contained in $G[Z_A]$ for otherwise we are done.

Since $L_A(v) \subseteq L'(v)$ for every $v \in V(A)$, c_A is an $L'|_{G[V(A)]}$ -coloring of $G[V(A)]$. We can extend c_A to be an L' -coloring c' of G by coloring each vertex v in $V(G) - V(A)$ with an arbitrary color in $L'(v)$. So M is a connected subgraph of $M'[V(M') \cap G[A]]$ for some monochromatic component M' with respect to c' intersecting $(Y'_1 \cap V(A)) \cup V(A \cap B)$. Since $G[Y'_1 \cap V(A)] \subseteq G[Z_A]$, if $V(M) \cap (Y_1 \cup Z) \neq \emptyset$, then M' intersects $Y_1 \cup Z$, so M' is contained in $G[Y'_1]$ by Lemma 22 (since L' is a (Z, ℓ) -growth of L), and hence $V(M) \subseteq Y'_1 \cap V(A) \subseteq Z_A$, a contradiction. So $V(M) \cap (Y_1 \cup Z) = \emptyset$. Hence $V(M) \cap ((Y'_1 \cap V(A) - (Y_1 \cup Z)) \cup (V(A \cap B) - Y'_1)) \neq \emptyset$.

Note that $(Y'_1 \cap V(A) - (Y_1 \cup Z)) \cup (V(A \cap B) - Y'_1) \subseteq U_A^{(0)} \subseteq Z_A$. So M intersects $U_A^{(0)} \subseteq Z_A$ and $V(M) \not\subseteq Z_A$. Hence there exist $v \in V(M) - Z_A$ and $y \in V(M) \cap Z_A$ such that $y \in N_{G[V(A)]}(v) \cap Z_A$ and $L''_A(v) \cap L''_A(y) \neq \emptyset$. Since $Z_A \subseteq Y''_A$ and L''_A is an (s, Y''_A, ℓ, r) -list-assignment, (R3) and (R4) imply that the vertices of M are colored with some color in $[s+2]$. So $V(M) \cap Z_A \subseteq V(G'')$.

Since M intersects $U_A^{(0)}$, the definition of $L'_A(v)$ implies that M intersects $U_A^{(1)}$. In particular, M contains a vertex in $Z_A - V(B)$. Moreover, since L_A is a (Y''_A, ℓ) -growth of L''_A and M intersects $U_A^{(0)} \subseteq Y''_A$, M is contained in $G[Y''_A] \subseteq G[Y_A]$ by Lemma 22.

Hence, to prove this claim, it suffices to show that some monochromatic component of G^* respect to $c_{\mathcal{L}}$ contains $M[V(M) \cap Z_A]$.

Since $Y'_A = Z_A$ and $L'_A(v) \cap L'_A(y) \supseteq L''_A(v) \cap L''_A(y) \neq \emptyset$, the definition of L'_A implies that $v \in W_A$. By the definition of $L'_{A,W}$, M does not have color k_A . Note that every component of $M[V(M) \cap Z_A]$ is contained in some monochromatic component of G^* with respect to $c_{\mathcal{L}}$ and either intersects $U_A^{(0)}$ or intersects $U_A^{(1)}$. By the definition of L'_A , every component of $M[V(M) \cap Z_A]$ intersects $U_A^{(1)}$. Since M does not have color k_A , by Claim 23.7, some monochromatic component of G^* with respect to $c_{\mathcal{L}}$ contains $M[V(M) \cap Z_A]$. This proves the claim. $\square$

For every $(A, B) \in \mathcal{L}$, let L_A and Y_A be the list-assignment and set mentioned in Claim 23.8, respectively, so $|Y_A| \leq \eta^*(\theta)$. For every $(A, B) \in \mathcal{L}$, since $Y_1 \cap V(A) \subseteq Y_A \subseteq V(A)$, there exists an (η, g) -bounded L_A -coloring c_A of $G[V(A)]$ such that for every $x \in [s+3, s+2+r]$, the set of vertices colored x is a stable set by our assumption. Since $\mathcal{S}$ is a segregation of $G-Z$ and $\mathcal{L}$ is a location, for every $v \in V(G) - V(G')$, there uniquely exists $(A_v, B_v) \in \mathcal{L}$ such that $v \in V(A_v) - V(B_v)$. Let c^* be the function, where $c^*(v) := c_{\mathcal{L}}(v)$ for every $v \in V(G')$, and $c^*(v) := c_{A_v}(v)$ for every $v \in V(G) - V(G')$. Clearly, c^* is an L' -coloring (and hence an L -coloring) of G . Note that for every $v \in V(A) \cap V(G')$ for some $(A, B) \in \mathcal{L}$, $c^*(v) = c_{\mathcal{L}}(v) = c_{A_v}(v)$ by the property of L_A .

Suppose that there exists $x^* \in [s+3, s+2+r]$ such that the set of vertices colored x^* is a not stable set. Then there exists an edge e of G whose both ends are colored x^* . Since for every $(A, B) \in \mathcal{L}$, the set of vertices colored x^* under c_A is a stable set, e belongs to $\bigcap_{(A,B) \in \mathcal{L}} B \subseteq G'$. But the set of vertices colored x^* under $c_{\mathcal{L}}$ is a stable set by Claim 23.7, a contradiction.

So c^* is not (η, g) -bounded by our assumption. By Lemma 22, the union U^* of the monochromatic components with respect to c^* intersecting $Y_1 \cup Z$ is contained in $G[Y'_1]$. Note that $|Y_1| > 3\theta \geq 1$. So U^* contains at most $|Y'_1| \leq h(|Y_1| + \xi_0) \leq g^*(|Y_1|) \leq g(|Y_1|) \leq |Y_1|^2 g(|Y_1|)$ by Lemma 22. Note that $|Y_1| \leq \eta$. Hence, there exists a monochromatic component M with respect to c^* disjoint from $Y_1 \cup Z$ containing more than $\eta^2 g(\eta)$ vertices.

Claim 23.9. *There exists a monochromatic component of G^* with respect to $c_{\mathcal{L}}$ containing $M[V(M) \cap V(G')]$.*

Proof. Let P be a path in M between two vertices x, y in $V(M) \cap V(G')$ internally disjoint from $V(M) \cap V(G')$ having at least one internal vertex. To prove this claim, it suffices to show that there exists a monochromatic component of G^* with respect to $c_{\mathcal{L}}$ containing both x and y .

Since $G' \supseteq G[\bigcap_{(A,B) \in \mathcal{L}} V(B)]$, there exists $(A, B) \in \mathcal{L}$ such that $P \subseteq A$. Let M_x and M_y be the components of $M[V(M) \cap V(A) \cap V(G')]$ containing x and y , respectively. By the existence of P , there exists a component M_P of $M[V(M) \cap V(A)]$ containing M_x and M_y , and M_P is not contained in $G[Z_A]$. Hence $V(M) \not\subseteq V(A)$, for otherwise $M = M_P$ is a monochromatic component with respect to c_A and hence contains at most $\eta^2 g(\eta)$ vertices, a contradiction. So M_P is a monochromatic component with respect to

c_A intersecting $(Y'_1 \cap V(A)) \cup V(A \cap B)$ but not contained in $G[Z_A]$. By Claim 23.8, some monochromatic component of G^* with respect to $c_{\mathcal{L}}$ contains $M_P[V(M_P) \cap Z_A]$ and hence contains both x and y . $\square$

For every $(A, B) \in \mathcal{L}$, let S_A be the set of the components of $M[V(M) \cap V(A)]$. Let $\mathcal{L}^* = \{(A, B) \in \mathcal{L} : V(M) - V(B) \neq \emptyset\}$.

Claim 23.10. *For every $(A, B) \in \mathcal{L}^*$, $V(M) \cap Z_A - V(B) \neq \emptyset$ and $\sum_{C \in S_A} |V(C)| \leq \eta^*(\theta)$.*

Proof. Let $(A, B) \in \mathcal{L}^*$. Since c_A is (η, g) -bounded, $V(M) \not\subseteq V(A)$, so each member of S_A is a monochromatic component of $G[A]$ with respect to c_A intersecting $(Y'_1 \cap V(A)) \cup V(A \cap B)$. So by Claim 23.8, for every member C of S_A , either C is contained in $G[Z_A]$, or C contains a vertex in $Z_A - V(B)$ and is contained in $G[Y_A]$. Hence if some member of S_A satisfies the latter, then $V(M) \cap Z_A - V(B) \neq \emptyset$; if every member of S_A satisfies the former, then $V(M) \cap V(A) \subseteq Z_A$, so $V(M) \cap Z_A - V(B) = V(M) - V(B) \neq \emptyset$ since $(A, B) \in \mathcal{L}^*$. Therefore, $V(M) \cap Z_A - V(B) \neq \emptyset$. Moreover, since $Z_A \subseteq Y_A$, $\sum_{C \in S_A} |V(C)| \leq |Y_A| \leq \eta^*(\theta)$. $\square$

By Claims 23.7 and 23.9, $|V(M) \cap \bigcap_{(A, B) \in \mathcal{L}} V(B)| \leq |V(M \cap G')| \leq \eta_1(\eta + \theta)$. By Claim 23.10, $|\mathcal{L}^*| \leq |V(M \cap G')| \leq \eta_1(\eta + \theta)$, and

$$\begin{aligned} |V(M)| &= |V(M) \cap \bigcap_{(A, B) \in \mathcal{L}} V(B)| + \sum_{(A, B) \in \mathcal{L}^*, C \in S_A} |V(C) - V(B)| \\ &\leq \eta_1(\eta + \theta) + |\mathcal{L}^*| \cdot \eta^*(\theta) \\ &\leq \eta_1(\eta + \theta) + \eta_1(\eta + \theta) \cdot \eta^*(\theta) \\ &= \eta_1(\eta + \theta) \cdot (1 + \eta^*(\theta)) \\ &\leq \eta_1(2\eta) \cdot (1 + \eta^*(\eta)) \\ &\leq g^*(\eta) \leq \eta^2 g(\eta), \end{aligned}$$

a contradiction. This completes the proof. $\square$

We now prove our main theorems. Before proving Theorem 24, we note that it implies our main results from Section 1. In particular, the first part of Theorem 24 with $Y_1 = \emptyset$ implies Theorem 2 since the list-assignment $(L(v) : v \in V(G))$ with $L(v) = [s + 2]$ for every $v \in V(G)$ is an $(s, \emptyset, 0, 0)$ -list-assignment. The second part of Theorem 24 with $Y_1 = \emptyset$ and $\ell = 1$ implies Theorem 7 since the list-assignment $(L(v) : v \in V(G))$ with $L(v) = [2s + 1]$ for every $v \in V(G)$ is an $(s, \emptyset, 1, s - 1)$ -list-assignment.

Theorem 24. *For all $s, t \in \mathbb{N}$ and for every graph H , there exists $\eta \in \mathbb{N}$ and a nondecreasing function g such that the following hold:*

1. If G is a graph with no $K_{s,t}$ subgraph and no H -minor, $Y_1 \subseteq V(G)$ with $|Y_1| \leq \eta$, and L is an $(s, Y_1, 0, 0)$ -list-assignment of G , then there exists an (η, g) -bounded L -coloring.
2. If G is a graph with no $K_{s,t}$ subgraph and no odd H -minor, $Y_1 \subseteq V(G)$ with $|Y_1| \leq \eta$, $\ell \in [s+2]$ and L is an $(s, Y_1, \ell, s-1)$ -list-assignment of G , then there exists an (η, g) -bounded L -coloring such that for every $x \in [s+3, 2s+1]$, the set of vertices colored x is a stable set in G .

Proof. Define the following:

- Let f be the function $f_{s,t}$ in Lemma 11.
- Let $f_0 : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ be the identity function, and for every $i \in \mathbb{N}$, let $f_i : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ be the function defined by $f_i(x) := f_{i-1}(x) + f(f_{i-1}(x))$.
- Let C_0 be the integer c in Theorem 19.
- Let $t' := \lceil (C_0 + 9)|V(H)|\sqrt{\log 12|V(H)|} \rceil$.
- Let θ_0 be the number θ^* and let g_0, η_0 be the functions g^*, η^* , respectively, in Lemma 23 taking $s = s, t = t$ and $t' = t'$.
- Let $\xi := 8|V(H)|$.
- Let $h : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ be the function mentioned in Lemma 22 by taking $s = s$ and $t = t$.
- Let $\theta' := \theta_0 + t' + 1$.
- Let η_1, g_1 be the number η and the function g , respectively, mentioned in Lemma 13 by taking $s = s, t = t$ and $k = 9\theta'$.
- Let $\eta_3 := h(\theta' + h(4\theta') + 1 + f(h(4\theta') + 1))$.
- Define $\eta := \eta_0(\theta') + \eta_1 + f(\eta_1) + h(4\theta') + 1 + f(h(4\theta') + 1) + \eta_3$.
- Let $\eta_2 := h(\eta + \xi) + f(h(\eta + \xi))$.
- Define $g : \mathbb{N} \rightarrow \mathbb{N}$ be the function defined by $g(x) := x + g_0(x) + g_1(x) + \eta_1 + \eta_2 \cdot \eta_3$ for every $x \in \mathbb{N}$.

Suppose to the contrary that there exists a graph G with no $K_{s,t}$ subgraph, a subset Y_1 of $V(G)$ with $|Y_1| \leq \eta$, a number $\ell \in [s+2]$, and list-assignment L of G such that:

- If G has no H -minor, then L is an $(s, Y_1, 0, 0)$ -list-assignment such that there exists no (η, g) -bounded L -coloring of G .
- If G has no odd H -minor, then L is an $(s, Y_1, \ell, s-1)$ -list-assignment of G such that there exists no (η, g) -bounded L -coloring c of G such that $\{v \in V(G) : c(v) = x\}$ is a stable set in G for every $x \in [s+3, 2s+1]$.

We further choose G and Y_1 so that $|V(G)|$ is as small as possible, and subject to this, $|Y_1|$ is as large as possible. Let $r := 0$ and $\ell' := 0$ when G has no H -minor; let $r := s-1$ and $\ell' := \ell$ when G has no odd H -minor.

Claim 24.1. $Y_1 \neq \emptyset$ and $N_G(Y_1) \neq \emptyset$.

Proof. If $Y_1 = \emptyset$, then let v be a vertex of G and define L' to be a $(\{v\}, \{\ell\} \cup [s+3, 2s+1])$ -progress of L . Let $Y'_1 = \{v\}$. By Lemma 12, L' is an $(s, Y'_1, 0, 0)$ -list-assignment when G has no H -minor; L' is an $(s, Y'_1, \ell, s-1)$ -list-assignment when G has no odd H -minor. In addition, $|Y'_1| \leq \eta$. So the maximality of Y_1 implies that there exists an (η, g) -bounded L' -coloring c' of G such that if G has no odd H -minor, then $\{v \in V(G) : c'(v) = x\}$ is a stable set in G for every $x \in [s+3, 2s+1]$. But c' is an (η, g) -bounded L -coloring c of G such that if G has no odd H -minor, then $\{v \in V(G) : c(v) = x\}$ is a stable set in G for every $x \in [s+3, 2s+1]$, a contradiction.

So $Y_1 \neq \emptyset$. Suppose that $N_G(Y_1) = \emptyset$. Let $G' = G - Y_1$. Then $L|_{G'}$ is an $(s, \emptyset, 0, 0)$ -list-assignment of G' when G has no H -minor; $L|_{G'}$ is an $(s, \emptyset, \ell, s-1)$ -list-assignment of G' when G has no odd H -minor. By the minimality of G , there exists an (η, g) -bounded $L|_{G'}$ -coloring c of G' such that if G has no odd H -minor, then $\{v \in V(G') : c(v) = x\}$ is a stable set in G' for every $x \in [s+3, 2s+1]$. By further coloring each vertex y in Y_1 with the unique element in $L(y)$, since $|Y_1| \leq |Y_1|^2 g(|Y_1|)$, there exists an (η, g) -bounded L -coloring of G such that if G has no odd H -minor, then $\{v \in V(G') : c(v) = x\}$ is a stable set in G' for every $x \in [s+3, 2s+1]$, a contradiction. This proves the claim. $\square$

Claim 24.2. $|Y_1| \geq 9\theta' + 1$.

Proof. Suppose $|Y_1| \leq 9\theta'$. So $|Y_1| < \eta_1$.

Since $\eta \geq \eta_1$ and $g \geq g_1$, if there exists an (η_1, g_1) -bounded L -coloring of G such that for every $x \in [s+3, 2s+1]$, the set of vertices colored x is a stable set in G , then it is an (η, g) -bounded L -coloring of G such that for every $x \in [s+3, 2s+1]$, the set of vertices colored x is a stable set in G , a contradiction. So there exists no (η_1, g_1) -bounded L -coloring of G such that for every $x \in [s+3, 2s+1]$, the set of vertices colored x is a stable set in G .

Hence, by Lemma 13 (with taking $F = [s+3, 2s+1]$) and Claim 24.1, there exist $Y'_1 \subseteq V(G)$ with $|Y_1| < |Y'_1| \leq \eta_1$ and an $(s, r+2, Y'_1)$ -list-assignment L' of G with $L'(v) \subseteq L(v)$ for every $v \in V(G)$ such that:

- there exists no (η_1, g_1) -bounded L' -coloring of G such that for every $x \in [s+3, 2s+1]$, the set of vertices colored x is a stable set in G ,
- for every L' -coloring of G , every monochromatic component intersecting Y_1 is contained in $G[Y'_1]$,
- $\{y \in Y_1 : L(y) \cap [s+3, 2s+1] \neq \emptyset\} = \{y \in Y'_1 : L'(y) \cap [s+3, 2s+1] \neq \emptyset\}$,
- for every $x \in \{\ell\} \cup [s+3, 2s+1]$ and $y \in Y'_1$ with $x \in L'(y)$, we have $\{v \in N_G(y) - Y'_1 : x \in L'(v)\} = \emptyset$, and
- for every $v \in V(G) - Y'_1$, we have $L'(v) \cap [s+3, 2s+1] = L(v) \cap [s+3, 2s+1]$.

So L' is an $(s, Y'_1, 0, 0)$ -list-assignment of G when G has no H -minor; L' is an $(s, Y'_1, \ell, s-1)$ -list-assignment of G when G has no odd H -minor.

Since $\eta \geq |Y'_1| > |Y_1|$, the maximality of Y_1 implies that there exists an (η, g) -bounded L' -coloring c' of G such that for every $x \in [s+3, 2s+1]$, the set of vertices colored x is a stable set in G . So every monochromatic component with respect to c' contains at most $\eta^2 g(\eta)$ vertices. Since c' is an L' -coloring, every monochromatic component with respect to c' intersecting Y_1 is contained in $G[Y'_1]$ and hence contains at most $|Y'_1| \leq \eta_1 \leq |Y_1|^2 g(|Y_1|)$ vertices. Since $L'(v) \subseteq L(v)$ for every $v \in V(G)$, c' is an L -coloring. Therefore, c' is an (η, g) -bounded L -coloring of G such that for every $x \in [s+3, 2s+1]$, the set of vertices colored x is a stable set in G , a contradiction. $\square$

Define $\mathcal{T} = \{(A, B) : |V(A \cap B)| < \theta', |V(A) \cap Y_1| \leq 3\theta'\}$ to be a set of separations of G .

Claim 24.3. $\mathcal{T}$ is a tangle in G of order θ' .

Proof. Suppose that $\mathcal{T}$ is not a tangle in G of order θ' . Since G has no $K_{s,t}$ subgraph and L is an $(s, r+2, Y_1)$ -list-assignment of G with $\eta \geq |Y_1| \geq 9\theta' + 1$ by Claim 24.2. By Lemma 16 by taking $s = s, t = t, \theta = \theta', \eta = \eta, g = g, r = r+2$ and $F = \{\ell'\} \cup [s+3, 2s+1]$, there exists an induced subgraph G' of G with $|V(G')| < |V(G)|$, a subset $Y'_1 \subseteq V(G')$ with $|Y'_1| \leq \eta$, and an $(s, r+2, Y'_1)$ -list-assignment L' of G' with $L'(v) \subseteq L(v)$ for every $v \in V(G)$ such that:

- there exists no (η, g) -bounded L' -coloring of G' such that for every $x \in [s+3, 2s+1]$, the set of vertices colored x is a stable set,
- $\{v \in Y'_1 : (\{\ell'\} \cup [s+3, 2s+1]) \cap L'(v) \neq \emptyset\} = \{v \in Y_1 : (\{\ell'\} \cup [s+3, 2s+1]) \cap L(v) \neq \emptyset\} \cap V(G')$, and
- for every $v \in V(G') - Y'_1$, we have $L'(v) \cap (\{\ell'\} \cup [s+3, 2s+1]) = L(v) \cap (\{\ell'\} \cup [s+3, 2s+1])$.

Hence L' is an (s, Y_1, ℓ', r) -list-assignment of G' . This contradicts the minimality of G . $\square$

Claim 24.4. $\mathcal{T}$ controls a $K_{t'}$ -minor.

Proof. Suppose to the contrary that $\mathcal{T}$ does not control a $K_{t'}$ minor. Note that $\theta' \geq \theta_0$, $\eta \geq \eta_0(\theta')$ and $g \geq g_0$. By Lemma 23, since there does not exist an (η, g) -bounded L -coloring of G such that for every $x \in [s+3, 2s+1]$, the set of vertices color with x is a stable set in G , we know there exist $(A^*, B^*) \in \mathcal{T}$, a set Y_{A^*} with $|Y_{A^*}| \leq \eta_0(\theta') \leq \eta$ and $Y_1 \cap V(A^*) \subseteq Y_{A^*} \subseteq V(A^*)$, and an (s, Y_{A^*}, ℓ', r) -list-assignment L_{A^*} of $G[V(A^*)]$ such that there exists no (η, g) -bounded L_{A^*} -coloring of $G[V(A^*)]$ such that for every $x \in [s+3, 2s+1]$, the set of vertices colored x is a stable set in $G[V(A^*)]$. But $|V(A^*)| < |V(G)|$ since $|V(A^*) \cap Y_1| \leq 3\theta' < |Y_1|$. So it contradicts the minimality of G . $\square$

By Claim 24.4, G contains a $K_{t'}$ -minor α . In particular, G has no odd H -minor and L is an $(s, Y_1, \ell, s-1)$ -list-assignment of G .

By Theorem 19, either G contains an odd $K_{|V(H)|}$ -minor, or there exists a set $Z \subseteq V(G)$ with $|Z| < 8|V(H)| \leq \xi$ such that the unique block U of $G - Z$ intersecting all branch vertices of α disjoint from Z is bipartite. The former implies that G contains an odd H -minor, a contradiction. So the latter holds.

Let (A_0, B_0) be the separation of G with $V(A_0 \cap B_0) = Z$ and $U \subseteq B_0$, and subject to this, $|V(A_0)|$ is maximal, and subject to these, $|E(A_0)|$ is minimal. Note that $A_0 - Z$ is the union of all components of $G - Z$ disjoint from U . For each cut-vertex v of $G - Z$ contained in U , let (A_v, B_v) be the separation of G with $V(A_v \cap B_v) = Z \cup \{v\}$ and $A_0 \cup U \subseteq B_v$, and subject to this, $|V(A_v)|$ is maximal, and subject to these, $|E(A_v)|$ is minimal. Define $\mathcal{L} = \{(A_0, B_0), (A_v, B_v) : v \text{ is a cut-vertex of } G - Z \text{ contained in } U\}$. Then $\mathcal{L}$ is a location by the maximality of $V(A_0)$ and $V(A_v)$ and the minimality of $|E(A_0)|$ and $|E(A_v)|$. Note that for every $(A, B) \in \mathcal{L}$, the order of (A, B) is at most $|Z| + 1 \leq \xi < t' < \theta'$, so either $(A, B) \in \mathcal{T}$ or $(B, A) \in \mathcal{T}$ by Claim 24.3.

Claim 24.5. $\mathcal{L} \subseteq \mathcal{T}$.

Proof. Suppose that $\mathcal{L} \not\subseteq \mathcal{T}$. Let $(A, B) \in \mathcal{L} - \mathcal{T}$. Then either $(A, B) = (A_0, B_0)$ and $(B_0, A_0) \in \mathcal{T}$, or there exists a cut-vertex v of $G - Z$ contained in U such that $(A, B) = (A_v, B_v)$ and $(B_v, A_v) \in \mathcal{T}$. Since $t' \geq |Z| + 2$, there exist vertices u_1, u_2 of H such that Q_1, Q_2 are disjoint from Z , where Q_1, Q_2 are the branch sets of α corresponding to u_1, u_2 , respectively. By the definition of U , U intersects both Q_1 and Q_2 . Since $Q_1 \cap Q_2 = \emptyset$ and $U \subseteq V(B)$, one of Q_1, Q_2 is contained in $B - V(A)$. Since $\mathcal{T}$ controls α and $(B, A) \in \mathcal{T}$ has order less than t' , both Q_1 and Q_2 intersect $V(A) - V(B)$, a contradiction. Hence $\mathcal{L} \subseteq \mathcal{T}$. $\square$

Let L' be a (Z, ℓ) -growth of L , and let $Y'_1 = \{v \in V(G) : |L'(v)| = 1\}$. By Lemma 22, $|Y'_1| \leq h(|Y_1 \cup Z|) \leq h(\eta + \xi)$ and for every $(A, B) \in \mathcal{L}$, we have $|Y'_1 \cap V(A)| \leq h(|A \cap B| + |Y_1 \cap V(A)|) \leq h(4\theta')$.

Claim 24.6. For every $v \in V(G) - Y'_1$, we have $L'(v) \cap (\{\ell\} \cup [s + 3, 2s + 1]) \neq \emptyset$ and $L'(v) - (\{\ell\} \cup [s + 3, 2s + 1]) \neq \emptyset$.

Proof. By Lemma 22, L' is an $(s, Y'_1, \ell, s-1)$ -list-assignment of G . So L' is an $(s, s+1, Y'_1)$ -list-assignment of G . By (L5), $|L'(v)| \geq s + 2$ for every $v \in V(G) - Y'_1$. Since $|\{\ell\} \cup [s + 3, 2s + 1]| = s$, we have $L'(v) - (\{\ell\} \cup [s + 3, 2s + 1]) \neq \emptyset$ for every $v \in V(G) - Y'_1$. Since $|[2s + 1] - (\{\ell\} \cup [s + 3, 2s + 1])| = s + 1$, we have $L'(v) \cap (\{\ell\} \cup [s + 3, 2s + 1]) \neq \emptyset$ for every $v \in V(G) - Y'_1$. $\square$

Let $G' = G[(\bigcap_{(A,B) \in \mathcal{L}} V(B)) \cup Y'_1]$. Note that $G' = G[V(U) \cup Y'_1]$. Observe that $G' - Y'_1 \subseteq U$, so $G' - Y'_1$ is bipartite. Let $\{P, Q\}$ be a bipartition of $G' - Y'_1$. Define an $L'|_{G'}$ -coloring c' of G' as follows.

- If $v \in Y'_1$, then let $c'(v)$ be the unique element of $L'(v)$.

- If $v \in P$, then let $c'(v)$ be an element of $L'(v) \cap (\{\ell\} \cup [s+3, 2s+1])$.
- If $v \in Q$, then let $c'(v)$ be an element of $L'(v) - (\{\ell\} \cup [s+3, 2s+1])$.

Note that c' is well-defined by Claim 24.6.

Claim 24.7. *Every monochromatic component with respect to c' contains at most η_2 vertices. Furthermore, for every $x \in [s+3, 2s+1]$, the set of vertices colored x is a stable set in G' .*

Proof. Let $x \in [2s+1]$, and let M be a monochromatic component with respect to c' such that all vertices of M are colored x . If $V(M) \cap Y'_1 = \emptyset$, then $V(M) \subseteq P$ or $V(M) \subseteq Q$, so M consists of one vertex as P and Q are stable sets in G' . So we may assume that $V(M) \cap Y'_1 \neq \emptyset$.

Since L' is an $(s, Y'_1, \ell, s-1)$ -list-assignment of G , by (R3), if $x \in \{\ell\} \cup [s+3, 2s+1]$, then either $V(M) \cap Y'_1 = \emptyset$ or $V(M) \subseteq Y'_1$. Recall $V(M) \cap Y'_1 \neq \emptyset$, so if $x \in \{\ell\} \cup [s+3, 2s+1]$, then $V(M) \subseteq Y'_1$. Hence, if $x \in [s+3, 2s+1]$, then $V(M) \subseteq Y'_1$, so M consists of one vertex by (R4); if $x = \ell$, then $V(M) \subseteq Y'_1$, so M contains at most $|Y'_1| \leq h(\eta + \xi) \leq \eta_2$ vertices.

So we may assume that $x \in [s+2] - \{\ell\}$. In particular, $V(M) \cap P = \emptyset$. Since L' is an $(s, Y'_1, \ell, s-1)$ -list-assignment, L' is an $(s, s+1, Y'_1)$ -list-assignment, so $|N_G(v) \cap Y'_1| \geq s$ for every vertex $v \in V(M) - Y'_1$ with $N_G(v) \cap Y'_1 \cap V(M) \neq \emptyset$ by (L3). By Lemma 11, $|\{v \in V(M) - Y'_1 : N_G(v) \cap Y'_1 \cap V(M) \neq \emptyset\}| \leq f(|Y'_1|)$. Note that every component of $M - Y'_1$ is contained in $G'[Q]$ which is a graph with no edge. Since $V(M) \cap Y'_1 \neq \emptyset$, every vertex in $V(M) - Y'_1$ has a neighbor in $Y'_1 \cap V(M)$. So $\{v \in V(M) - Y'_1 : N_G(v) \cap Y'_1 \cap V(M) \neq \emptyset\} = V(M) - Y'_1$. Hence $|V(M)| \leq |V(M) \cap Y'_1| + |V(M) - Y'_1| \leq |Y'_1| + f(|Y'_1|) \leq h(\eta + \xi) + f(h(\eta + \xi)) = \eta_2$. $\square$

Let $Y''_1 := V(G')$. Note that $Y''_1 \supseteq Y'_1$. Let L'' be the following list-assignment of G :

- For every $v \in Y'_1$, let $L''(v) := L'(v)$.
- For every $v \in V(G') - Y'_1$, let $L''(v) := \{c'(v)\}$.
- For every $v \in N_G^{<s}(Y''_1)$, let $L''(v)$ be a subset of $L'(v) - \{L''(y) : y \in Y''_1 \cap N_G(v)\}$ of size $|L'(v)| - |N_G(v) \cap (Y''_1 - Y'_1)|$ such that $|(L'(v) - L''(v)) \cap [s+3, 2s+1]| = |\{y \in N_G(v) \cap Y''_1 - Y'_1 : L''(y) \in [s+3, 2s+1]\}|$ if possible.
- For every $v \in V(G) - (Y''_1 \cup N_G^{<s}(Y''_1))$, let $L''(v) := L'(v)$.

Claim 24.8. *The following hold:*

- L'' is an $(s, s+1, Y''_1)$ -list-assignment.
- For every $x \in [s+3, 2s+1]$, $\{y \in Y''_1 : x \in L''(y)\} = \{y \in Y'_1 : x \in L'(y)\} \cup \{y \in V(G') - Y'_1 : c'(y) = x\}$ is a stable set in G .

- For every $v \in N_G^{<s}(Y_1'') \cup (V(G) - N_G[Y_1''])$, we have $(s-1) - |L''(v) \cap [s+3, 2s+1]| = |\{y \in N_G(v) \cap Y_1'' : L''(y) \subseteq [s+3, 2s+1]\}|$.
- For every $(A, B) \in \mathcal{L}$, we have $|Y_1'' \cap V(A)| \leq h(4\theta') + 1$.

Proof. Since L' is an $(s, s+1, Y_1')$ -list-assignment, L'' is an $(s, s+1, Y_1'')$ -list-assignment. And by Claim 24.7, for every $x \in [s+3, 2s+1]$,

$$\begin{aligned} \{y \in Y_1'' : x \in L''(y)\} &= \{y \in Y_1' : x \in L'(y)\} \cup \{y \in V(G') - Y_1' : c'(y) = x\} \\ &= \{y \in V(G') : c'(y) = x\} \end{aligned}$$

is a stable set in G .

Note that for every $v \in V(G) - Y_1''$, there exists $u \in U \cup \{0\}$ such that $v \in V(A_u) - V(B_u)$, so for every $y \in N_G(v) \cap V(G') - Y_1'$, $u \in U$ and y must be u , and hence $|N_G(v) \cap Y_1'' - Y_1'| \leq 1$.

For every $v \in N_G^{<s}(Y_1'')$, if $N_G(v) \cap Y_1'' = N_G(v) \cap Y_1'$, then $L''(v) \cap [s+3, 2s+1] = L'(v) \cap [s+3, 2s+1]$; if $N_G(v) \cap Y_1'' \neq N_G(v) \cap Y_1'$, then $|N_G(v) \cap Y_1'| \leq s-2$, and by (L3) and (L4), we know $|L'(v)| = 2s+1 - |N_G(v) \cap Y_1'| \geq s+3$, so $L'(v) \cap [s+2] \neq \emptyset \neq L'(v) \cap [s+3, 2s+1]$, and since $|N_G(v) \cap Y_1'' - Y_1'| \leq 1$, we know that $L''(v)$ can be chosen such that $|(L'(v) - L''(v)) \cap [s+3, 2s+1]| = |\{y \in N_G(v) \cap Y_1'' - Y_1' : L''(y) \in [s+3, 2s+1]\}|$.

Therefore, for every $v \in N_G^{<s}(Y_1'')$,

$$\begin{aligned} (s-1) - |L''(v) \cap [s+3, 2s+1]| &= (s-1) - |L'(v) \cap [s+3, 2s+1]| + |(L'(v) - L''(v)) \cap [s+3, 2s+1]| \\ &= |\{y \in N_G(v) \cap Y_1' : L'(y) \subseteq [s+3, 2s+1]\}| + |(L'(v) - L''(v)) \cap [s+3, 2s+1]| \\ &= |\{y \in N_G(v) \cap Y_1' : L'(y) \subseteq [s+3, 2s+1]\}| \\ &\quad + |\{y \in N_G(v) \cap Y_1'' - Y_1' : L''(y) \in [s+3, 2s+1]\}| \\ &= |\{y \in N_G(v) \cap Y_1'' : L''(y) \subseteq [s+3, 2s+1]\}|, \end{aligned}$$

where the second equality follows from the fact that L' satisfies (R5).

For every $v \in V(G) - N_G[Y_1'']$, since $L''(v) = L'(v)$, by (R5),

$$\begin{aligned} (s-1) - |L''(v) \cap [s+3, 2s+1]| &= (s-1) - |L'(v) \cap [s+3, 2s+1]| \\ &= |\{y \in N_G(v) \cap Y_1' : L'(y) \subseteq [s+3, 2s+1]\}| \\ &= 0 = |\{y \in N_G(v) \cap Y_1'' : L''(y) \subseteq [s+3, 2s+1]\}|. \end{aligned}$$

For every $(A, B) \in \mathcal{L}$, we have $|Y_1'' \cap V(A)| \leq |Y_1' \cap V(A)| + |(Y_1'' - Y_1') \cap V(A)| \leq h(4\theta') + |V(A \cap B) - Z| \leq h(4\theta') + 1$. $\square$

Let L''' be a $(N_G^{>s}(Y_1''), \{\ell\} \cup [s+3, 2s+1])$ -progress of L'' . Let $Y_1''' := \{v \in V(G) : |L'''(v)| = 1\}$. By Lemma 12 and Claim 24.8, L''' is an $(s, s+1, Y_1''')$ -list-assignment of G .

and satisfies (R3)-(R5). Hence L''' is an $(s, Y_1''', \ell, s-1)$ -list-assignment. Furthermore, for every $(A, B) \in \mathcal{L}$, since $V(A \cap B) \subseteq Y_1''$, we have $|Y_1''' \cap V(A)| \leq |Y_1'' \cap V(A)| + f(|Y_1'' \cap V(A)|)$ by Lemma 11, so $|Y_1''' \cap V(A)| \leq h(4\theta') + 1 + f(h(4\theta') + 1) \leq \eta$ by Claim 24.8.

Let L^* be an $(\emptyset, \ell)$ -growth of L''' , and let $Y_1^* := \{v \in V(G) : |L^*(v)| = 1\}$. By Lemma 22, L^* is an $(s, Y_1^*, \ell, s-1)$ -list-assignment of G such that:

- For every L^* -coloring of G , every monochromatic component intersecting Y_1''' is contained in $G[Y_1^*]$.
- For every $(A, B) \in \mathcal{L}$, we have $|Y_1^* \cap V(A)| \leq h(|V(A \cap B)| + |Y_1''' \cap V(A)|) \leq h(\theta' + h(4\theta') + 1 + f(h(4\theta') + 1)) \leq \eta_3 \leq \eta$.

For every $(A, B) \in \mathcal{L}$, since $V(A \cap B) \subseteq Y_1^*$, we have $L^*|_{G[V(A)]}$ is an $(s, Y_1^* \cap V(A), \ell, s-1)$ -list-assignment of $G[V(A)]$ such that $|\{y \in V(A) : |L^*(y)| = 1\}| = |Y_1^* \cap V(A)| \leq \eta_3 \leq \eta$. For every $(A, B) \in \mathcal{L}$, since $(A, B) \in \mathcal{T}$, we have $|V(A)| < |V(G)|$. Therefore, for every $(A, B) \in \mathcal{L}$, by the minimality of G , there exists an (η, g) -bounded $L^*|_{G[V(A)]}$ -coloring c_A^* of $G[V(A)]$ such that $\{v \in V(A) : c_A^*(v) = x\}$ is a stable set in $G[V(A)]$ for every $x \in [s+3, 2s+1]$.

Since $\bigcap_{(A,B) \in \mathcal{L}} V(B) \subseteq V(G') \subseteq Y_1^*$ and $\mathcal{L}$ is a location, for every $v \in V(G) - Y_1^*$, there uniquely exists $(A_v, B_v) \in \mathcal{L} \subseteq \mathcal{T}$ such that $v \in V(A_v) - V(B_v)$. Let c^* be the following function:

- For every $v \in Y_1^*$, let $c^*(v)$ be the unique element in $L^*(v)$.
- For every $v \in V(G) - Y_1^*$, let $c^*(v) := c_{A_v}^*(v)$.

Clearly, c^* is well-defined and is an L^* -coloring of G .

Suppose that there exists $x^* \in [s+3, 2s+1]$ such that the set $\{v \in V(G) : c^*(v) = x^*\}$ is not a stable set in G . Then there exists an edge e of G such that both ends of e are colored x^* under c^* . If there exists $(A, B) \in \mathcal{L}$ such that e belongs to A , then $\{v \in V(A) : c_A^*(v) = x^*\}$ is not a stable set in $G[V(A)]$, a contradiction. So $e \in \bigcap_{(A,B) \in \mathcal{L}} B \subseteq G'$, and hence $\{v \in V(G') : c'(v) = x^*\}$ is not a stable set in G' , contradicting Claim 24.7.

This shows that for every $x \in [s+3, 2s+1]$, the set $\{v \in V(G) : c^*(v) = x\}$ is a stable set in G . Hence c^* is not (η, g) -bounded.

Since $L^*(v) \subseteq L'(v)$ for every $v \in V(G)$, c^* is an L' -coloring of G . Since L' is an (Z, ℓ) -growth of L , by Lemma 22, every monochromatic component with respect to c^* intersecting $Y_1 \cup Z$ is contained in $G[Y_1']$. So the union of monochromatic components with respect to c^* intersecting $Y_1 \cup Z$ is contained in $G[Y_1']$ and hence contains at most $|Y_1'| \leq h(\eta + \xi) \leq \eta_2 \leq g(0) \leq g(|Y_1|) \leq g(\eta)$. In particular, the union of monochromatic components with respect to c^* intersecting Y_1 contains at most $|Y_1|^2 g(|Y_1|)$ vertices.

Hence there exists a monochromatic component M with respect to c^* disjoint from $Y_1 \cup Z$ such that M contains more than $\eta^2 g(\eta)$ vertices. Suppose that $V(M) \cap V(G') \neq \emptyset$. Since $V(M) \cap Z = \emptyset$ and $|V(A \cap B) - Z| \leq 1$ for every $(A, B) \in \mathcal{L}$, $G[V(M) \cap V(G')]$ is connected. Since $c^*(v) = c'(v)$ for every $v \in V(G')$, $M \cap G'$ is a monochromatic

component with respect to c' , so $|V(M \cap G')| \leq \eta_2$ by Claim 24.7. For every $(A, B) \in \mathcal{L}$, if $V(M) - V(B) \neq \emptyset$, then $V(M) \cap V(A \cap B) - Z \neq \emptyset$ since $V(M)$ is disjoint from Z . By the definition of $\mathcal{L}$, if $(A_1, B_1), (A_2, B_2)$ are distinct members of $\mathcal{L}$, then $V(A_1 \cap B_1) - Z \neq V(A_2 \cap B_2) - Z$. Let $\mathcal{L}' = \{(A, B) \in \mathcal{L} : V(M) \cap V(A \cap B) - Z \neq \emptyset\}$. So $|\mathcal{L}'| \leq |V(M) \cap V(G')| \leq \eta_2$. For every $(A, B) \in \mathcal{L}'$, since $V(M)$ intersects $V(A \cap B) - Z \subseteq Y_1''' \cap V(A)$, $G[V(M) \cap V(A)]$ is a monochromatic component with respect to c_A^* intersecting $Y_1''' \cap V(A) \subseteq Y_1^* \cap V(A)$, so by Lemma 22, $G[V(M) \cap V(A)]$ is contained in $G[Y_1^* \cap V(A)]$ and hence contains at most $|Y_1^* \cap V(A)| \leq \eta_3$ vertices. Therefore, $|V(M)| \leq |V(M) \cap V(G')| \cdot \eta_3 \leq \eta_2 \eta_3 \leq g(0) \leq \eta^2 g(\eta)$, a contradiction.

Hence $V(M) \cap V(G') = \emptyset$. In particular, $V(M) - V(B) \neq \emptyset$ for some $(A, B) \in \mathcal{L}$. For every $(A', B') \in \mathcal{L}$, since $V(A' \cap B') \subseteq \bigcap_{(A, B) \in \mathcal{L}} V(B) \subseteq V(G')$, $V(M)$ is disjoint from $V(A' \cap B')$. So if there exist $(A_1, B_1), (A_2, B_2) \in \mathcal{L}$ such that $V(M) - V(B_1) \neq \emptyset \neq V(M) - V(B_2)$, then since $\mathcal{L}$ is a location and M is connected, $V(M)$ intersects $V(A_1 \cap B_1)$, a contradiction. So there uniquely exists $(A^*, B^*) \in \mathcal{L}$ such that $V(M) - V(B^*) \neq \emptyset$. Since $V(M)$ is disjoint from $V(A^* \cap B^*)$, we have $V(M) \subseteq V(A^*) - V(B^*)$. Hence M is a monochromatic component with respect to $c_{A^*}^*$. Since $c_{A^*}^*$ is an (η, g) -bounded $L^*|_{G[V(A^*)]}$ -coloring, M contains at most $\eta^2 g(\eta)$ vertices, a contradiction. This proves the lemma. $\square$

6. About the Gerards-Seymour conjecture

This section proves Theorem 6. The proof depends on the following result of Kang and Oum [25]. For graphs G and H , let $G + H$ be the graph obtained from the disjoint union of G and H by adding all edges with one end in $V(G)$ and one end in $V(H)$. Let $K_{s,t}^* := K_s + I_t$, where I_t is the graph on t vertices with no edges.

Lemma 25 (Corollary of [25, Lemma 5.1]). *Let $t, d \in \mathbb{N}$ with $d \geq 3$. If there exists a positive integer $\eta \geq 4t - 3$ such that every odd K_{t+1} -minor-free graph containing no bipartite $K_{2t,t+1}^*$ -subdivision has a d -coloring with clustering η , then every odd K_{t+1} -minor-free graph has a $(d + 4t - 3)$ -coloring with clustering η .*

Proof. Let $\mathcal{F}$ be the set of all graphs whose every component contains at most η vertices. So $\mathcal{F}$ is a class of graphs closed under isomorphism and taking disjoint union, and $\mathcal{F}$ contains all graphs on at most $4t - 3$ vertices. Since every graph with no odd K_{t+1} -minor and no bipartite $K_{2t,t+1}^*$ -subdivision has a d -coloring with clustering η , by [25, Lemma 5.1], every odd K_{t+1} -minor-free graph has a $(d + 4t - 3)$ -coloring with clustering η . $\square$

We now prove Theorem 6.

Theorem 26. *For every $s \in \mathbb{N}$, there exists $\eta \in \mathbb{N}$ such that every odd K_{s+1} -minor-free graph has an $(8s - 4)$ -coloring with clustering η .*

Proof. By Theorem 7, there exists η such that every odd K_{s+1} -minor-free graph with no $K_{2s-1, \binom{2s-1}{2}+s+2}$ subgraph has a $(2(2s-1)+1)$ -coloring with clustering η . So every odd K_{s+1} -minor-free graph with no bipartite $K_{2s, s+1}^*$ -subdivision has a $(4s-1)$ -coloring with clustering η . We may assume that $\eta \geq 4s-3$. The theorem now follows from Lemma 25. $\square$

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